AN OPENSOURCE EBOOK

INTRODUCTION TO SQL



Bobby Iliev

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About the book

- **This version was published on October 13,2021**

This open-source introduction to SQL guide will help you learn the basics of SQL and start using relational databases for your SysOps, DevOps, and Dev projects. Whether you are a DevOps/SysOps engineer, developer, or just a Linux enthusiast, you will most likely have to use SQL at some point in your career.

The guide is suitable for anyone working as a developer, system administrator, or DevOps engineer who wants to learn the basics of SQL.

About the author

My name is Bobby Iliev, and I have been working as a Linux DevOps Engineer since 2014. I am an avid Linux lover and supporter of the open-source movement philosophy. I am always doing that which I cannot do in order that I may learn how to do it, and I believe in sharing knowledge.

I think it's essential always to keep professional and surround yourself with good people, work hard, and be nice to everyone. You have to perform at a consistently higher level than others. That's the mark of a true professional.

For more information, please visit my blog at https://bobbyiliev.com, follow me on Twitter @bobbyiliev_ and YouTube.

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Databases

Before we dive deep into SQL, let's quickly define what a database is.

The definition of databases from Wikipedia is:

A database is an organized collection of data, generally stored and accessed electronically from a computer system.

In other words, a database is a collection of data stored and structured in different database tables.

Tables and columns

You've most likely worked with spreadsheet systems like Excel or Google Sheets. At the very basic, database tables are quite similar to spreadsheets.

Each table has different **columns** which could contain different types of data.

For example, if you have a todo list app, you would have a database, and in your database, you would have different tables storing different information like:

- Users - In the users table, you would have some data for your users like: `username`, `name`, and `active`, for example.
- Tasks - The tasks table would store all of the tasks that you are planning to do. The columns of the tasks table would be for example, `task_name`, `status`, `due_date` and `priority`.

The Users table will look like this:

```
+----+----------+---------------+--------+
| id | username | name          | active |
+----+----------+---------------+--------+
| 1  |    bobby | Bobby Iliev   |   true |
| 2  |    grisi | Greisi I.     |   true |
| 3  |  devdojo | Dev Dojo      |  false |
+----+----------+---------------+--------+
```

Rundown of the table structure:

- We have 4 columns: `id`, `username`, `name` and `active`.
- We also have 3 entries/users.
- The `id` column is a unique identifier of each user and is auto-

incremented.

In the next chapter, we will learn how to install MySQL and create our first database.

MySQL

Now that you know what a database, table, and column are, the next thing that you would need to do is install a database service where you would be running your SQL queries on.

We will be using MySQL as it is free, open-source, and very widely used.

Installing MySQL

Depending on your operating system, to install MySQL run the following commands.

Install MySQL on Ubuntu

To install MySQL on a Linux or Ubuntu machine, run the following commands:

- First update your `apt` repository:

```
sudo apt update -y
```

- Then install MySQL:

```
sudo apt install mysql-server mysql-client
```

We are installing two packages, one is the actual MySQL server, and the other is the MySQL client, which would allow us to connect to the MySQL server and run our queries.

To check if MySQL is running, run the following command:

```
sudo systemctl status mysql.service
```

To secure your MySQL server, you could run the following command:

```
sudo mysql_secure_installation
```

Then follow the prompt and choose a secure password and save it in a secure place like a password manager.

With that, you would have MySQL installed on your Ubuntu server. The above should also work just fine on Debian.

Install MySQL on Mac

I would recommend installing MySQL using Homebrew:

```
brew install mysql
```

After that, start MySQL:

```
brew services start mysql
```

And finally, secure it:

```
mysql_secure_installation
```

In case that you ever need to stop the MySQL service, you could do so with the following command:

```
brew services stop mysql
```

Install MySQL on Windows

To install MySQL on Windows, I would recommend following the steps from the official documentation here:

https://dev.mysql.com/doc/refman/8.0/en/windows-installation.html

Accessing MySQL via CLI

To access MySQL run the `mysql` command followed by your user:

```
mysql -u root -p
```

Creating a database

After that, switch to the `demo` database that we created in the previous chapter:

```
USE demo;
```

To exit the just type the following:

```
exit;
```

Configuring `.my.cnf`

By configuring the `~/.my.cnf` file in your user's home directory, MySQL would allow you to log in without prompting you for a password.

To make that change, what you need to do is first create a `.my.cnf` file in your user's home directory:

```
touch ~/.my.cnf
```

After that, set secure permissions so that other regular users could not read the file:

```
chmod 600 ~/.my.cnf
```

Then using your favourite text editor, open the file:

```
nano ~/.my.cnf
```

And add the following configuration:

```
[client]
user=YOUR_MYSQL_USERNAME
password=YOUR_MYSQL_PASSWORD
```

Make sure to update your MySQL credentials accordingly, then save the file and exit.

After that, if you run just `mysql`, you will be authenticated directly with the credentials that you've specified in the `~/.my.cnf` file without being

prompted for a password.

The mysqladmin command

As a quick test, you could check all of your open SQL connections by running the following command:

```
mysqladmin proc
```

The `mysqladmin` tool would also use the client details from the `~/.my.cnf` file, and it would list your current MySQL process list.

Another cool thing that you could try doing is combining this with the `watch` command and kind of monitor your MySQL connections in almost real-time:

```
watch -n1 mysqladmin proc
```

To stop the `watch` command, just hit `CTRL+C`

GUI clients

If you prefer using GUI clients, you could take a look a the following ones and install them locally on your laptop:

- MySQL Workbench
- Sequel Pro
- TablePlus

This will allow you to connect to your database via a graphical interface rather than the `mysql` command-line tool.

If you want to have a production-ready MySQL database, I would recommend giving DigitalOcean a try:

Worry-free managed database hosting

Tables

Before we get started with SQL, let's learn how to create tables and columns.

As an example, we are going to create a `users` table with the following columns:

- `id` - this is going to be the primary key of the table and would be the unique identifier of each user.
- `username` - this column would hold the username of our users.
- `name` - here, we will store the full name of users.
- `status` - here, we will store the status of a user, which would indicate if a user is active or not.

You need to specify the data type of each column.

In our case it would be like this:

- `id` - Integer
- `username` - Varchar
- `name` - Varchar
- `status` - Number

Data types

The most common data types that you would come across are:

- `CHAR`(size): Fixed-length character string with a maximum length of 255 bytes.
- `VARCHAR`(size): Variable-length character string. Max size is specified in parenthesis.
- `TEXT`(size): A string with a maximum length of 65,535 bytes.
- `INTEGER`(size) or `INT`(size): A medium integer.
- `BOOLEAN` or `BOOL`: Holds a true or false value.
- `DATE`: Holds a date.

Let's have the following users table as an example:

- `id`: We would want to set the ID to `INT`.
- `name`: The name should fit in a `VARCHAR` column.
- `about`: As the about section could be longer, we could set the column data type to `TEXT`.
- `birthday`: For the birthday column of the user, we could use `DATE`.

For more information on all data types available, make sure to check out the official documentation here.

Creating a database

As we briefly covered in the previous chapter, before you could create tables, you would need to create a database by running the following:

- First access MySQL:

```
mysql -u root -p
```

- Then create a database called `demo_db`:

```
CREATE DATABASE demo_db;
```

Note: the database name needs to be unique, if you already have a database named `demo_db` you would receive an error that the database already exists.

You can consider this database as the container where we would create all of the tables in.

Once you've created the database, you need to switch to that database:

```
USE demo_db;
```

You can think of this as accessing a directory in Linux with the `cd` command. With `USE`, we switch to a specific database.

Alternatively, if you do not want to 'switch' to the specific database, you would need to specify the so-called fully qualified table name. For example, if you had a `users` table in the `demo_db`, and you wanted to

select all of the entries from that table, you could use one of the following two approaches:

- Switch to the `demo_db` first and then run a select statement:

```
USE demo_db;
SELECT username FROM users;
```

- Alternatively, rather than using the `USE` command first, specify the database name followed by the table name separated with a dot: `db_name.table_name`:

```
SELECT username FROM demo_db.users;
```

We are going to cover the `SELECT` statement more in-depth in the following chapters.

Creating tables

In order to create a table, you need to use the `CREATE TABLE` statement followed by the columns that you want to have in that table and their data type.

Let's say that we wanted to create a `users` table with the following columns:

- `id`: An integer value
- `username`: A varchar value
- `about`: A text type
- `birthday`: Date
- `active`: True or false

The query that we would need to run to create that table would be:

```
CREATE TABLE users
(
    id INT,
    username VARCHAR(255),
    about TEXT,
    birthday DATE,
    active BOOL
);
```

Note: You need to select a database first with the `USE` command as mentioned above. Otherwise you will get the following error: `ERROR 1046 (3D000): No database selected.

To list the available tables, you could run the following command:

```
SHOW TABLES;
```

Output:

```
+-------------------+
| Tables_in_demo_db |
+-------------------+
| users             |
+-------------------+
```

Creating a new table from an existing table

You can create a new table from an existing table by using the `CREATE TABLE AS` statement.

Let's test that by creating a new table from the table `users` which we created earlier.

```
CREATE TABLE users2 AS
(
    SELECT * FROM users
);
```

The output that you would get would be:

```
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Note: When creating a table in this way, the new table will be populated with the records from the existing table (based on the SELECT Statement)

Rename tables

You can rename a table by using `ALTER TABLE` statement.

Let's change name of user2 table to user3

```
ALTER TABLE user2 RENAME TO user3
```

Dropping tables

You can drop or delete tables by using the `DROP TABLE` statement.

Let's test that and drop the table that we've just created:

```
DROP TABLE users;
```

The output that you would get would be:

```
Query OK, 0 rows affected (0.03 sec)
```

And now, if you were to run the `SHOW TABLES;` query again, you would get the following output:

```
Empty set (0.00 sec)
```

Allowing NULL values

By default, each column in your table can hold NULL values. In case that you don't wanted to allow NULL values for some of the columns in a specific table, you need to specify this during the table creation or later on change the table to allow that.

For example, let's say that we want the `username` column to be a required one, we would need to alter the table create statement and include `NOT NULL` right next to the `username` column like this:

```
CREATE TABLE users
(
    id INT,
    username VARCHAR(255) NOT NULL,
    about TEXT,
    birthday DATE,
    active BOOL
);
```

That way, when you try to add a new user, MySQL will let you know that the `username` column is required.

Specifying a primary key

The primary key column, which in our case is the `id` column, is a unique identifier for our users.

We want the `id` column to be unique, and also, whenever we add new users, we want the ID of the user to autoincrement for each new user.

This can be achieved with a primary key and `AUTO_INCREMENT`. The primary key column needs to be `NOT NULL` as well.

If we were to alter the table creation statement, it would look like this:

```
CREATE TABLE users
(
    id INT PRIMARY KEY AUTO_INCREMENT,
    username VARCHAR(255) NOT NULL,
    about TEXT,
    birthday DATE,
    active BOOL
);
```

Index Optimization for Database Queries

In database management, establishing a PRIMARY KEY for tables is fundamental. Using our previous example, the `id` column serves as this primary key. However, as the volume of data grows, searching by attributes other than the primary key, like the `date of birth`, can become increasingly slow. To optimize such queries, you can introduce an INDEX on specific columns.

Consider the `birthday` column. To enhance the speed of queries focused on this column, an INDEX can be pivotal:

```
CREATE INDEX birthday_idx ON users(birthday);
```

Tip: For queries spanning multiple fields, you have the option to create a composite index incorporating all the relevant fields. Let's say, for example, you want to index both the `birthday` and `active` columns.

The order in which you list the fields in a composite index matters. For instance, given that the `active` column might have limited unique values (e.g., true or false) compared to the `birthday` column, the sequence of fields in the index can influence efficiency. Designing the index like this:

```
CREATE INDEX users_multi_idx ON users(active, birthday);
```

May not be as efficient as:

```
CREATE INDEX users_multi_idx ON users(birthday, active);
```

Placing `birthday` first in the index ensures a quicker reduction in potential matches, optimizing the server's data manipulation process.

Updating tables

In the above example, we created a new table and then dropped it as it was empty. However, in a real-life scenario, this would really be the case.

So whenever you need to add or remove a new column from a specific table, you would need to use the `ALTER TABLE` statement.

Let's say that we wanted to add an `email` column with type varchar to our `users` table.

The syntax would be:

```
ALTER TABLE users ADD email VARCHAR(255);
```

After that, if you were to describe the table, you would see the new column:

```
DESCRIBE users;
```

Output:

```
+----------+--------------+------+-----+---------+
| Field    | Type         | Null | Key | Default |
+----------+--------------+------+-----+---------+
| id       | int          | NO   | PRI | NULL    |
| username | varchar(255) | NO   |     | NULL    |
| about    | text         | YES  |     | NULL    |
| birthday | date         | YES  |     | NULL    |
| active   | tinyint(1)   | YES  |     | NULL    |
| email    | varchar(255) | YES  |     | NULL    |
+----------+--------------+------+-----+---------+
```

If you wanted to drop a specific column, the syntax would be:

```
ALTER TABLE table_name DROP COLUMN column_name;
```

Note: Keep in mind that this is a permanent change, and if you have any critical data in the specific column, it would be deleted instantly.

You can use the `ALTER TABLE` statement to also change the data type of a specific column. For example, you could change the `about` column from `TEXT` to `LONGTEXT` type, which could hold longer strings.

Note: Important thing to keep in mind is that if a specific table already holds a particular type of data value like an integer, you can't alter it to varchar, for example. Only if the column does not contain any values, then you could make the change.

Truncate table

The `TRUNCATE TABLE` command is used to **delete all of the data** from an existing table, but not the table itself.

- Syntax of Truncate table:

```
TRUNCATE TABLE table_name;
```

- Example:

Consider a Sellers table having the following records:

```
+----+----------+-----+-----------+----------+
| ID | NAME     |Items| CITY      | SALARY   |
+----+----------+-----+-----------+----------+
|  1 | Shivam   |  34 | Ahmedabad |  2000.00 |
|  2 | Ajay     |  22 | Delhi     |  4400.00 |
|  3 | Kaushik  |  28 | Kota      |  2000.00 |
|  4 | Chaitali |  25 | Mumbai    |  6600.00 |
|  5 | Hardik   |  26 | Bhopal    |  8100.00 |
|  6 | Maria    |  23 | MP        |  4200.00 |
|  7 | Muffy    |  29 | Indore    |  9000.00 |
+----+----------+-----+-----------+----------+
```

Following is the example of a Truncate command:

```
TRUNCATE TABLE Sellers;
```

After that if you do a `COUNT(*)` on that table you would see that the table is completely empty.

Basic Syntax

In this chapter, we will go over the basic SQL syntax.

SQL statements are basically the 'commands' that you run against a specific database. Through the SQL statements, you are telling MySQL what you want it to do, for example, if you wanted to get the `username` of all of your users stored in the `users` table, you would run the following SQL statement:

```
SELECT username FROM users;
```

Rundown of the statement:

- `SELECT`: First, we specify the `SELECT` keyword, which indicates that we want to select some data from the database. Other popular keywords are: `INSERT`, `UPDATE` and `DELETE`.
- `username`: Then we specify which column we want to select.
- `FROM users`: After that, we specify the table that we want to select the data from using the `FROM` keyword.
- The semicolon `;` is highly recommended to put at the end. Standard SQL syntax requires it, but some "Database Management Systems' (DBMS)" are tolerant about it, but it's not worth the risk.

If you run the above statement, you will get no results as the new `users` table that we've just created is empty.

As a good practice, all SQL keywords should be with uppercase, however, it would work just fine if you use lower case as well.
Let's go ahead and cover the basic operations next.

INSERT

To add data to your database, you would use the `INSERT` statement.

Let's use the table that we created in the last chapter and insert 1 user into our `users` table:

```
INSERT INTO users (username, email, active)
VALUES ('bobby', 'bobby@bobbyiliev.com', true);
```

Rundown of the insert statement:

- `INSERT INTO`: first, we specify the `INSERT INTO` keyword, which tells MySQL that we want to insert data a table.
- `users (username, email, active)`: then, we specify the table name `users` and the columns that we want to insert data into.
- `VALUES`: then, we specify the values that we want to insert in. The order of attributes is the same as in `users (...)`.

SELECT

Once we've inserted that user, let's go ahead and retrieve the information.

To retrieve information from your database, you could use the `SELECT` statement:

```
SELECT * FROM users;
```

Output:

```
+----+----------+-------+----------+--------+---------------+
| id | username | about | birthday | active | email         |
+----+----------+-------+----------+--------+---------------+
|  1 | bobby    | NULL  | NULL     |      1 | bobby@b...com |
+----+----------+-------+----------+--------+---------------+
```

We specify `*` right after the `SELECT` keyword, this means that we want to get all of the columns from the `users` table.

If we wanted to retrieve only the `username` and the `email` columns instead, we would change the statement to:

```
SELECT username, email FROM users;
```

This will return all of the users, but as of the time being we have only 1:

```
+----------+----------------------+
| username | email                |
+----------+----------------------+
| bobby    | bobby@bobbyiliev.com |
+----------+----------------------+
```

UPDATE

In order to modify data in your database, you could use the `UPDATE` statement.

The syntax would look like this:

```
UPDATE users SET username='bobbyiliev' WHERE id=1;
```

Rundown of the statement:

- `UPDATE users`: First, we specify the `UPDATE` keyword followed by the table that we want to update.
- `SET username='bobbyiliev'`: Then we specify the columns that we want to update and the new value that we want to set.
- `WHERE id=1`: Finally, by using the `WHERE` clause, we specify which user should be updated. In our case it is the user with ID 1.

NOTE: If we don't specify a `WHERE` clause, all of the entries inside the `users` table would be updated, and all users would have the `username` set to `bobbyiliev`. You need to be careful when you use the `UPDATE` statement without a `WHERE` clause, as every single row will be updated.

We are going to cover `WHERE` more in-depth in the next few chapters.

DELETE

As the name suggests, the `DELETE` statement would remove data from your database.

The syntax is as follows:

```
DELETE FROM users WHERE id=1;
```

Similar to the `UPDATE` statement, if you don't specify a `WHERE` clause, all of the entries from the table will be affected, meaning that all of your users will be deleted.

Comments

In case that you are writing a larger SQL script, it might be helpful to add some comments so that later on, when you come back to the script, you would know what each line does.

As with all programming languages, you can add comments in SQL as well.

There are two types of comments:

- Inline comments:

To do so, you just need to add `--` before the text that you want to comment out:

```
SELECT * FROM users; -- Get all users
```

- Multiple-line comments:

Similar to some other programming languages in order to comment multiple lines, you could wrap the text in `/* */` as follows:

```
/*
Get all of the users
from your database
*/
SELECT * FROM users;
```

You could write that in a `.sql` file and then run it later on, or execute the few lines directly.

Conclusion

Those were some of the most common basic SQL statements.

In the next chapters, we are going to go over each of the above statements more in-depth.

SELECT

As we briefly covered in the previous chapter, the `SELECT` statement allows us to retrieve data from single or multiple tables on the database. In this chapter, we will be performing the query on a single table.

It corresponds to the projection operation of Relational Algebra.

You can use `SELECT` to get all of your users or a list of users that match a certain criteria.

Before we dive into the `SELECT` statement let's quickly create a database:

```
CREATE DATABASE sql_demo;
```

Switch to that database:

```
USE sql_demo;
```

Create a new users table:

```
CREATE TABLE users
(
    id INT PRIMARY KEY AUTO_INCREMENT,
    username VARCHAR(255) NOT NULL,
    about TEXT,
    email VARCHAR(255),
    birthday DATE,
    active BOOL
);
```

Insert some data that we could work with:

```
INSERT INTO users
  ( username, email, active )
VALUES
  ('bobby', 'b@devdojo.com', true),
  ('devdojo', 'd@devdojo.com', false),
  ('tony', 't@devdojo.com', true);
```

Output:

```
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

We are going to learn more about the `INSERT` statement in the following chapters.

SELECT all columns

Now that we've got some data in the `users` table, let's go ahead and retrieve all of the entries from that table:

```
SELECT * FROM users;
```

Rundown of the statement:

- `SELECT`: First, we specify the action that we want to execute, in our case, we want to select or get some data from the database.
- `*`: The star here indicates that we want to get all of the columns associated with the table that we are selecting from.
- `FROM`: The from statement tells MySQL which table we want to select the data from. You need to keep in mind that you can select from multiple tables, but this is a bit more advanced, and we are going to cover this in the next few chapters.
- `users`: This is the table name that we want to select the data from.

This will return all of the entries in the `users` table along with all of the columns:

```
+----+----------+-------+----------+--------+---------------+
| id | username | about | birthday | active | email         |
+----+----------+-------+----------+--------+---------------+
|  1 | bobby    | NULL  | NULL     |      1 | b@devdojo.com |
|  2 | devdojo  | NULL  | NULL     |      0 | d@devdojo.com |
|  3 | tony     | NULL  | NULL     |      1 | t@devdojo.com |
+----+----------+-------+----------+--------+---------------+
3 rows in set (0.00 sec)
```

As you can see, we get a list of the 3 users that we've just created, including all of the columns in that table. In some cases, the table might

have a lot of columns, and you might not want to see all of them. For example, we have the `about` and `birthday` columns that are all `NULL` at the moment. So let's see how we could limit that and get only a list of specific columns.

Pattern matching

SQL pattern matching let's you to search for patterns if you don't know the exact word or phrase you are looking for. To do this, we use so-called wildcard characters to match a pattern together with LIKE and ILIKE operators.

Two of the most common wildcard characters are `_` and `%`.

`_` matches any single character and `%` matches an arbitrary number of characters.

Let's see an example how you would look for a `username` ending with `y`:

```
SELECT * FROM users WHERE username LIKE '%y';
```

Output:

```
+----+----------+-------+----------+--------+---------------+
| id | username | about | birthday | active | email         |
+----+----------+-------+----------+--------+---------------+
|  1 | bobby    | NULL  | NULL     |      1 | b@devdojo.com |
|  3 | tony     | NULL  | NULL     |      1 | t@devdojo.com |
+----+----------+-------+----------+--------+---------------+
```

As you can see above, we used `%` to match any number of characters preceding the character `y`.

If we know the exact number of characters we want to match, we can use `_`. Each `_` represents a single character.

So, if we want to look up an username that has `e` as its second character, we would do something like this:

```
SELECT * FROM users WHERE username LIKE '_e%';
```

Output:

```
+----+----------+-------+----------+--------+---------------+
| id | username | about | birthday | active | email         |
+----+----------+-------+----------+--------+---------------+
|  2 | devdojo  | NULL  | NULL     |      0 | d@devdojo.com |
+----+----------+-------+----------+--------+---------------+
```

Please, keep in mind that `LIKE` operator is case sensitive, meaning it won't mach capital letters with lowercase letters and vice versa. If you wish to ignore capitalization, use `ILIKE` operator instead.

Formatting

As we mentioned in the previous chapters, each SQL statement needs to end with a semi-colon: `;`. Alternatively, rather than using a semi-colon, you could use the `\G` characters which would format the output in a list rather than a table.

The syntax is absolutely the same but you just change the `;` with `\G`:

```
SELECT * FROM users \G
```

The output will be formatted like this:

```
*************************** 1. row ***************************
      id: 1
username: bobby
   about: NULL
birthday: NULL
  active: 1
   email: b@devdojo.com
*************************** 2. row ***************************
      id: 2
username: devdojo
   about: NULL
birthday: NULL
  active: 0
   email: d@devdojo.com
...
```

This is very handy whenever your table consists of a large number of columns and they can't fit on the screen, which makes it very hard to read the result set.

SELECT specific columns only

You could limit this to a specific set of columns. Let's say that you only needed the `username` and the `active` columns. In this case, you would change the `*` symbol with the columns that you want to select divided by a comma:

```
SELECT username,active FROM users;
```

Output:

```
+----------+--------+
| username | active |
+----------+--------+
| bobby    |      1 |
| devdojo  |      0 |
| tony     |      1 |
+----------+--------+
```

As you can see, we are getting back only the 2 columns that we've specified in the `SELECT` statement.

NOTE: *SQL names are case insensitive. For example, username ≡ USERNAME ≡ userName.*

SELECT with no FROM Clause

In a SQL statement, a column can be a literal with no `FROM` clause.

```
SELECT 'Sunil' as username;
```

Output:

```
+----------+
| username |
+----------+
| Sunil    |
+----------+
```

SELECT with Arithmetic Operations

The select clause can contain arithmetic expressions involving the operation +, –, *, and /.

```
SELECT username, active*5 as new_active FROM users;
```

Output:

```
+----------+------------+
| username | new_active |
+----------+------------+
| bobby    |          5 |
| devdojo  |          0 |
| tony     |          5 |
+----------+------------+
```

LIMIT

The `LIMIT` clause is very handy in case that you want to limit the number of results that you get back. For example, at the moment, we have 3 users in our database, but let's say that you only wanted to get 1 entry back when you run the `SELECT` statement.

This can be achieved by adding the `LIMIT` clause at the end of your statement, followed by the number of entries that you want to get. For example, let's say that we wanted to get only 1 entry back. We would run the following query:

```
SELECT * FROM users LIMIT 1;
```

Output:

```
+----+----------+-------+----------+--------+---------------+
| id | username | about | birthday | active | email         |
+----+----------+-------+----------+--------+---------------+
|  2 | bobby    | NULL  | NULL     |      1 | b@devdojo.com |
+----+----------+-------+----------+--------+---------------+
```

If you wanted to get 2 entries, you would change `LIMIT 2` and so on.

COUNT

In case that you wanted to get only the number of entries in a specific column, you could use the `COUNT` function. This is a function that I personally use very often.

The syntax is the following:

```
SELECT COUNT(*) FROM users;
```

Output:

```
+----------+
| COUNT(*) |
+----------+
|        3 |
+----------+
```

MIN, MAX, AVG, and SUM

Another useful set of functions similar to `COUNT` that would make your life easier are:

- `MIN`: This would give you the smallest value of a specific column. For example, if you had an online shop and you wanted to get the lowest price, you would use the `MIN` function. In our case, if we wanted to get the lowest user ID, we would run the following:

```
SELECT MIN(id) FROM users;
```

This would return `1` as the lowest user ID that we have is 1.

- `MAX`: Just like `MIN`, but it would return the highest value:

```
SELECT MAX(id) FROM users;
```

In our case, this would be `3` as we have only 3 users, and the highest value of the `id` column is 3.

- `AVG`: As the name suggests, it would sum up all of the values of a specific column and return the average value. As we have 3 users with ids 1, 2, and 3, the average would be 6 divided by 3 users which is 2.

```
SELECT AVG(id) FROM users;
```

- `SUM`: This function takes all of the values from the specified column and sums them up:

```
SELECT SUM(id) FROM users;
```

DISTINCT

In some cases, you might have duplicate entries in a table, and in order to get only the unique values, you could use `DISTINCT`.

To better demonstrate this, let's run the insert statement one more time so that we could duplicate the existing users and have 6 users in the users table:

```
INSERT INTO users
  ( username, email, active )
VALUES
  ('bobby', 'b@devdojo.com', true),
  ('devdojo', 'd@devdojo.com', false),
  ('tony', 't@devdojo.com', true);
```

Now, if you run `SELECT COUNT(*) FROM users;` you would get `6` back.

Let's also select all users and show only the `username` column:

```
SELECT username FROM users;
```

Output:

```
+----------+
| username |
+----------+
| bobby    |
| devdojo  |
| tony     |
| bobby    |
| devdojo  |
| tony     |
+----------+
```

As you can see, each name is present multiple times in the list. We have `bobby`, `devdjo` and `tony` showing up twice.

If we wanted to show only the unique `usernames`, we could add the `DISTINCT` keyword to our select statement:

```
SELECT DISTINCT username FROM users;
```

Output:

```
+----------+
| username |
+----------+
| bobby    |
| devdojo  |
| tony     |
+----------+
```

As you can see, the duplicate entries have been removed from the output.

Conclusion

The `SELECT` statement is essential whenever working with SQL. In the next chapter, we are going to learn how to use the `WHERE` clause and take the `SELECT` statements to the next level.

WHERE

The `WHERE` clause allows you to specify different conditions so that you could filter out the data and get a specific result set.

You would add the `WHERE` clause after the `FROM` clause.

The syntax would look like this:

```
SELECT column_name FROM table_name WHERE column=some_value;
```

WHERE Clause example

If we take the example `users` table from the last chapter, let's say that we wanted to get only the active users. The SQL statement would look like this:

```
SELECT DISTINCT username, email, activem FROM users WHERE
active=true;
```

Output:

```
+----------+---------------+--------+
| username | email         | active |
+----------+---------------+--------+
| bobby    | b@devdojo.com |      1 |
| tony     | t@devdojo.com |      1 |
+----------+---------------+--------+
```

As you can see, we are only getting `tony` and `bobby` back as their `active` column is `true` or `1`. If we wanted to get the inactive users, we would have to change the `WHERE` clause and set the `active` to `false`:

```
+----------+---------------+--------+
| username | email         | active |
+----------+---------------+--------+
| devdojo  | d@devdojo.com |      0 |
+----------+---------------+--------+
```

As another example, let's say that we wanted to select all users with the username `bobby`. The query, in this case, would be:

```
SELECT username, email, active FROM users WHERE
username='bobby';
```

The output would look like this:

```
+----------+---------------+--------+
| username | email         | active |
+----------+---------------+--------+
| bobby    | b@devdojo.com |      1 |
| bobby    | b@devdojo.com |      1 |
+----------+---------------+--------+
```

We are getting 2 entries back as we have 2 users in our database with the username `bobby`.

Operators

In the example, we used the `=` operator, which checks if the result set matches the value that we are looking for.

A list of popular operators are:

- `!=` : Not equal operator
- `>` : Greater than
- `>=` : Greater than or equal operator
- `<` : Less than operator
- `<=` : Less than or equal operator

For more information about other available operators, make sure to check the official documentation here.

AND keyword

In some cases, you might want to specify multiple criteria. For example, you might want to get all users that are active, and the username matches a specific value. This could be achieved with the `AND` keyword.

Syntax:

```
SELECT * FROM users WHERE username='bobby' AND active=true;
```

The result set would contain the data that matches both conditions. In our case, the output would be:

```
+----+----------+-------+----------+--------+---------------+
| id | username | about | birthday | active | email         |
+----+----------+-------+----------+--------+---------------+
|  2 | bobby    | NULL  | NULL     |      1 | b@devdojo.com |
|  5 | bobby    | NULL  | NULL     |      1 | b@devdojo.com |
+----+----------+-------+----------+--------+---------------+
```

If we were to change the `AND` statement to `active=false`, we would not get any results back as none of the entries in our database match that condition:

```
SELECT * FROM users WHERE username='bobby' AND active=false;
```

```
-- Output:
Empty set (0.01 sec)
```

OR keyword

In some cases, you might want to specify multiple criteria. For example, you might want to get all users that are active, or their username matches a specific value. This could be achieved with the `OR` keyword.

As with any other programming language, the main difference between `AND` and `OR` is that with `AND`, the result would only return the values that match the two conditions, and with `OR`, you would get a result that matches either of the conditions.

For example, if we were to run the same query as above but change the `AND` to `OR`, we would get all users that have the username `bobby` and also all users that are not active:

```
SELECT * FROM users WHERE username='bobby' OR active=false;
```

Output:

```
+----+----------+-------+----------+--------+---------------+
| id | username | about | birthday | active | email         |
+----+----------+-------+----------+--------+---------------+
|  2 | bobby    | NULL  | NULL     |      1 | b@devdojo.com |
|  3 | devdojo  | NULL  | NULL     |      0 | d@devdojo.com |
|  5 | bobby    | NULL  | NULL     |      1 | b@devdojo.com |
|  6 | devdojo  | NULL  | NULL     |      0 | d@devdojo.com |
+----+----------+-------+----------+--------+---------------+
```

LIKE operator

Unlike the `=` operator, the `LIKE` operator allows you to do wildcard matching similar to the `*` symbol in Linux.

For example, if you wanted to get all users that have the `y` letter in them, you would run the following:

```
SELECT * FROM users WHERE username LIKE '%y%';
```

Output

```
+----+----------+-------+----------+--------+---------------+
| id | username | about | birthday | active | email         |
+----+----------+-------+----------+--------+---------------+
|  2 | bobby    | NULL  | NULL     |      1 | b@devdojo.com |
|  4 | tony     | NULL  | NULL     |      1 | t@devdojo.com |
+----+----------+-------+----------+--------+---------------+
```

As you can see, we are getting only `tony` and `bobby` but not `devdojo` as there is no `y` in `devdojo`.

This is quite handy when you are building some search functionality for your application.

IN operator

The `IN` operator allows you to provide a list expression and would return the results that match that list of values.

For example, if you wanted to get all users that have the username `bobby` and `devdojo`, you could use the following:

```
SELECT * FROM users WHERE username IN ('bobby', 'devdojo');
```

Output:

```
+----+----------+-------+----------+--------+---------------+
| id | username | about | birthday | active | email         |
+----+----------+-------+----------+--------+---------------+
|  2 | bobby    | NULL  | NULL     |      1 | b@devdojo.com |
|  3 | devdojo  | NULL  | NULL     |      0 | d@devdojo.com |
|  5 | bobby    | NULL  | NULL     |      1 | b@devdojo.com |
|  6 | devdojo  | NULL  | NULL     |      0 | d@devdojo.com |
+----+----------+-------+----------+--------+---------------+
```

This allows you to simplify your `WHERE` expression so that you don't have to add numerous `OR` statements.

IS operator

If you were to run `SELECT * FROM users WHERE about=NULL;` you would get an empty result set as the `=` operator can't be used to check for NULL values. Instead, you would need to use the `IS` operator instead.

The `IS` operator is only used to check `NULL` values, and the syntax is the following:

```
SELECT * FROM users WHERE about IS NULL;
```

If you wanted to get the results where the value is not NULL, you just need to change `IS` to `IS NOT`:

```
SELECT * FROM users WHERE about IS NOT NULL;
```

BETWEEN operator

The `BETWEEN` operator allows to select value with a given range.The values can be numbers, text, or dates. BETWEEN operator is inclusive: begin and end values are included.

For Example if you want to select those user which have id between 3 and 6.

```
SELECT * FROM users WHERE id BETWEEN 3 AND 6;
```

Output:

```
+----+----------+-------+----------+--------+---------------+
| id | username | about | birthday | active | email         |
+----+----------+-------+----------+--------+---------------+
|  3 | devdojo  | NULL  | NULL     |      0 | d@devdojo.com |
|  5 | bobby    | NULL  | NULL     |      1 | b@devdojo.com |
|  6 | devdojo  | NULL  | NULL     |      0 | d@devdojo.com |
+----+----------+-------+----------+--------+---------------+
```

Conclusion

In this chapter, you've learned how to use the `WHERE` clause with different operators to get different type of results based on the parameters that you provide.

In the next chapter, we will learn how to order the result set.

Sorting with ORDER and GROUP BY

In the last chapter, you've learned how to use the `SELECT` statement with the `WHERE` clause and filter the result set based on some conditions.

More often than not, you would want to order the results in a specific way based on a particular column. For example, you might want to order the users alphabetically based on their username.

In this chapter, you will learn how to use the `ORDER BY` and `GROUP BY` clauses.

ORDER BY

The main thing that you need to keep in mind when using `ORDER BY` is to specify the column or columns you want to order by. In case you want to specify multiple columns to order by, you need to separate each column with a comma.

If we were to run the following statement without providing an `ORDER BY` clause:

```
SELECT id, username FROM users;
```

We will get the following output:

```
+----+----------+
| id | username |
+----+----------+
|  2 | bobby    |
|  3 | devdojo  |
|  4 | tony     |
|  5 | bobby    |
|  6 | devdojo  |
|  7 | tony     |
+----+----------+
```

As you can see, the result set is sorted by the primary key, which, in our case, is each user's id. If we wanted to sort the output by `username`, we would run the following query:

```
SELECT id, username FROM users ORDER BY username;
```

Note: The `ORDER BY` statement is followed by the column's name that we want to order by.

The output, in this case, will be:

```
+----+----------+
| id | username |
+----+----------+
|  2 | bobby    |
|  5 | bobby    |
|  3 | devdojo  |
|  6 | devdojo  |
|  4 | tony     |
|  7 | tony     |
+----+----------+
```

Note: You can use `ORDER BY` with and without specifying a `WHERE` clause. If you've used a `WHERE` clause, you must put the `ORDER BY` clause after the `WHERE` clause.

The default sorting is ascending and is specified with the `ASC` keyword, and you don't need to add it explicitly, but if you want to sort by descending order, you need to use the `DESC` keyword.

If we use the query above and add `DESC` at the end as follows:

```
SELECT id, username FROM users ORDER BY username DESC;
```

We will see the following output:

```
+----+----------+
| id | username |
+----+----------+
|  4 | tony     |
|  7 | tony     |
|  3 | devdojo  |
|  6 | devdojo  |
|  2 | bobby    |
|  5 | bobby    |
+----+----------+
```

As you can see, we've got the same list of users sorted alphabetically but in reverse order.

GROUP BY

The `GROUP BY` statement allows you to use a function like `COUNT`, `MIN`, `MAX` etc., with multiple columns.

For example, let's say that we wanted to get all user counts grouped by username.

In our case, we have two users with the username `bobby`, two users with the username `tony`, and two users with the username `devdojo`. This represented in an SQL statement would look like this:

```
SELECT COUNT(username), username FROM users GROUP BY username;
```

The output, in this case, would be:

```
+-----------------+----------+
| COUNT(username) | username |
+-----------------+----------+
|               2 | bobby    |
|               2 | devdojo  |
|               2 | tony     |
+-----------------+----------+
```

The `GROUP BY` statement grouped the identical usernames. Then it ran a `COUNT` on each of `bobby`, `tony` and `devdojo`.

The main thing to remember here is that the `GROUP BY` should be added after the `FROM` clause and after the `WHERE` clause.

HAVING Clause

The `HAVING` clause allows you to filter out the results on the groups formed by the `GROUP BY` clause.

For example, let's say that we wanted to get all usernames that are duplicates, i.e., all the usernames present in more than one table record.

In our case, we have two users with the username `bobby`, two users with the username `tony`, and two users with username `devdojo`. This represented in an SQL statement would look like this:

```
SELECT COUNT(username), username
FROM users
GROUP BY username
HAVING COUNT(username) > 1;
```

The output, in this case, would be:

```
+-----------------+----------+
| COUNT(username) | username |
+-----------------+----------+
|               2 | bobby    |
|               2 | devdojo  |
|               2 | tony     |
+-----------------+----------+
```

The `GROUP BY` clause grouped the identical usernames, calculated their counts and filtered out the groups using the `HAVING` clause.

NOTE:- *The WHERE clause places conditions on the selected columns, whereas the HAVING clause places conditions on groups created by the GROUP BY clause.*

INSERT

To add data to your database, you would use the `INSERT` statement. You can insert data into one table at a time only.

The syntax is the following:

```
INSERT INTO table_name
  (column_name_1,column_name_2,column_name_n)
VALUES
  ('value_1', 'value_2', 'value_3');
```

You would start with the `INSERT INTO` statement, followed by the table that you want to insert the data into. Then you would specify the list of the columns that you want to insert the data into. Finally, with the `VALUES` statement, you specify the data that you want to insert.

The important part is that you need to keep the order of the values based on the order of the columns that you've specified.

In the above example the `value_1` would go into `column_name_1`, the `value_2` would go into `column_name_2` and the `value_3` would go into `column_name_x`.

Let's use the table that we created in the last chapter and insert 1 user into our `users` table:

```
INSERT INTO users
  (username, email, active)
VALUES
  ('greisi', 'g@devdojo.com', true);
```

Rundown of the insert statement:

- `INSERT INTO users`: First, we specify the `INSERT INTO` keywords which tells MySQL that we want to insert data into the `users` table.
- `users (username, email, active)`: Then, we specify the table name `users` and the columns that we want to insert data into.
- `VALUES`: Then, we specify the values that we want to insert in.

Inserting multiple records

We've briefly covered this in one of the previous chapters, but in some cases, you might want to add multiple records in a specific table.

Let's say that we wanted to create 5 new users, rather than running 5 different queries like this:

```
INSERT INTO users (username, email, active) VALUES ('user1',
'user1@devdojo.com', true);
INSERT INTO users (username, email, active) VALUES ('user1',
'user2@devdojo.com', true);
INSERT INTO users (username, email, active) VALUES ('user1',
'user3@devdojo.com', true);
INSERT INTO users (username, email, active) VALUES ('user1',
'user4@devdojo.com', true);
INSERT INTO users (username, email, active) VALUES ('user1',
'user5@devdojo.com', true);
```

What you could do is to combine this into one `INSERT` statement by providing a list of the values that you want to insert as follows:

```
INSERT INTO users
  (username, email, active)
VALUES
  ('user1', 'user1@devdojo.com', true),
  ('user2', 'user2@devdojo.com', true),
  ('user3', 'user3@devdojo.com', true),
  ('user4', 'user4@devdojo.com', true),
  ('user5', 'user5@devdojo.com', true);
```

That way, you will add 5 new entries in your `users` table with a single `INSERT` statement. This is going to be much more efficient.

Inserting multiple records using another table

In the previous section, we have discussed how we can insert multiple records using a single INSERT query. But sometimes there are cases where we need to insert multiple records which are residing in some other table.

In this section, we are going to learn how we can insert multiple records at once using a single INSERT query.

Consider a table, say `prospect_users`, which stores the information of the people who want to become the users of our service, but they are not yet actual users.

In order to add them to our user database, we have to insert there entries into our `users` table. We can achieve the same by writing an `INSERT` query with multiple `VALUES` listed in them (as discussed in previous section).

But there is an easier way where we achieve the same by querying the `prospect_users` table.

```
INSERT INTO users (username, email, active)
SELECT username, email, active
FROM prospect_users
WHERE active=true;
```

Using the above statement, an entry for each active prospect users will be made in our `users` table.

UPDATE

As the name suggests, whenever you have to update some data in your database, you would use the `UPDATE` statement.

You can use the `UPDATE` statement to update multiple columns in a single table.

The syntax would look like this:

```
UPDATE users SET username='bobbyiliev' WHERE id=1;
```

Rundown of the statement:

- `UPDATE users`: First, we specify the `UPDATE` keyword followed by the table that we want to update.
- `username='bobbyiliev'`: Then we specify the columns that we want to update and the new value that we want to set.
- `WHERE id=1`: Finally, by using the `WHERE` clause, we specify which user should be updated. In our case, it is the user with ID 1.

The most important thing that you need to keep in mind is that if you don't specify a `WHERE` clause, all of the entries inside the `users` table would be updated, and all users would have the `username` set to `bobbyiliev`.

Important: You need to be careful when you use the `UPDATE` statement without a `WHERE` clause as every single row will be updated.

If you have been following along all of the user entries in our `users`

table, it currently have no data in the `about` column:

```
+----+----------+-------+
| id | username | about |
+----+----------+-------+
|  2 | bobby    | NULL  |
|  3 | devdojo  | NULL  |
|  4 | tony     | NULL  |
|  5 | bobby    | NULL  |
|  6 | devdojo  | NULL  |
|  7 | tony     | NULL  |
+----+----------+-------+
```

Let's go ahead and update this for all users and set the column value to `404 bio not found`, For example:

```
UPDATE users SET about='404 bio not found';
```

The output would let you know how many rows have been affected by the query:

```
Query OK, 6 rows affected (0.02 sec)
Rows matched: 6  Changed: 6  Warnings: 0
```

Now, if you were to run a select for all `users`, you would get the following result:

```
+----+----------+-------------------+
| id | username | about             |
+----+----------+-------------------+
|  2 | bobby    | 404 bio not found |
|  3 | devdojo  | 404 bio not found |
|  4 | tony     | 404 bio not found |
|  5 | bobby    | 404 bio not found |
|  6 | devdojo  | 404 bio not found |
|  7 | tony     | 404 bio not found |
+----+----------+-------------------+
```

Let's now say that we wanted to update the `about` column for the user with an id of 2. In this case, we need to specify a `WHERE` clause followed by the ID of the user that we want to update as follows:

```
UPDATE users SET about='Hello World :)' WHERE id=2;
```

The output here should indicate that only 1 row was updated:

```
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Now, if you again run the `SELECT id, username, about FROM users` query, you would see that the user with `id` of 2 now has an updated `about` column data:

```
+----+----------+--------------------+
| id | username | about              |
+----+----------+--------------------+
|  2 | bobby    | Hello World :)     |
|  3 | devdojo  | 404 bio not found |
|  4 | tony     | 404 bio not found |
|  5 | bobby    | 404 bio not found |
|  6 | devdojo  | 404 bio not found |
|  7 | tony     | 404 bio not found |
+----+----------+--------------------+
```

Updating records using another table

As we've seen in the previous section, you can insert multiple rows in your table using another table. You can use the same principle for the update command.

To do that you simply have to list all needed table in the `update` section, then you have to explain which action you want to perform on the table, and then you need to link the table together.

For example, if you want to update the `about` field in the `users` table using the content of the `about` field in the `prospect_users` table, you would do something like this:

```
update users, prospect_users
set users.about = prospect_users.about
where prospect_users.username = users.username;
```

DELETE

As the name suggests, the `DELETE` statement would remove data from your database.

The syntax is as follows:

```
DELETE FROM users WHERE id=5;
```

The output should indicate that 1 row was affected:

```
Query OK, 1 row affected (0.01 sec)
```

Important: Just like the `UPDATE` statement, if you don't specify a `WHERE` clause, all of the entries from the table will be affected, meaning that all of your users will be deleted. So, it is critical to always add a `WHERE` clause when executing a `DELETE` statement.

```
DELETE FROM users;
```

The output should indicate (where x is the number of tuples in the table):

```
Query OK, x row(s) affected (0.047 sec)
```

Similar to the Linux `rm` command, when you use the `DELETE` statement,

the data would be gone permanently, and the only way to recover your data would be by restoring a backup.

Delete from another table

As we saw in the two precedents sections you can `INSERT` or `UPDPATE` tables rows based on other table data. You can do the same for the `DELETE`.

For example, if you want to delete the records from the `users` table if the corresponding prospect has been disabled, you could do it this way:

```
delete users
from users, prospect_users
where users.username = prospect_users.username
and NOT prospect_users.active
```

JOIN

The `JOIN` clause allows you to combine the data from 2 or more tables into one result set.

As we will be selecting from multiple columns, we need to include the list of the columns we want to choose data from after the `FROM` clause is separated by a comma.

In this chapter, we will go over the following `JOIN` types:

- `CROSS` Join
- `INNER` Join
- `LEFT` Join
- `RIGHT` Join

Before we get started, let's create a new database and two tables that we are going to work with:

- We are going to call the database `demo_joins`:

```
CREATE DATABASE demo_joins;
```

- Then, switch to the new database:

```
USE demo_joins;
```

- Then, the first table will be called `users`, and it will only have two columns: `id` and `username`:

```
CREATE TABLE users
(
    id INT PRIMARY KEY AUTO_INCREMENT,
    username VARCHAR(255) NOT NULL
);
```

- Then, let's create a second table called `posts`, and to keep things simple, we will have three two columns: `id`, `user_id` and `title`:

```
CREATE TABLE posts
(
    id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT,
    title VARCHAR(255) NOT NULL
);
```

The `user_id` column would be used to reference the user's ID that the post belongs to. It is going to be a one to many relations, e.g. one user could have many posts:



- Now, let's add some data into the two tables first by creating a few users:

```
INSERT INTO users
  ( username )
VALUES
  ('bobby'),
  ('devdojo'),
  ('tony'),
  ('greisi');
```

- And finally add some posts:

```
INSERT INTO posts
  ( user_id, title )
VALUES
  ('1', 'Hello World!'),
  ('2', 'Getting started with SQL'),
  ('3', 'SQL is awesome'),
  ('2', 'MySQL is up!'),
  ('1', 'SQL - structured query language');
```

Now that we've got our tables and demo data ready, let's go ahead and learn how to use joins.

CROSS JOIN

The `CROSS` join allows you to put the result of two tables next to each other without specifying any `WHERE` conditions. This makes the `CROSS` join the simplest one, but it is also not of much use in a real-life scenario.

So if we were to select all of the users and all of the posts side by side, we would use the following query:

```
SELECT * FROM users CROSS JOIN posts;
```

The output will be all of your users and all of the posts side by side:

```
+----+----------+----+--------+-----------------+
| id | username | id |user_id | title           |
+----+----------+----+--------+-----------------+
|  4 | greisi   |  1 |      1 | Hello World!    |
|  3 | tony     |  1 |      1 | Hello World!    |
|  2 | devdojo  |  1 |      1 | Hello World!    |
|  1 | bobby    |  1 |      1 | Hello World!    |
|  4 | greisi   |  2 |      2 | Getting started |
|  3 | tony     |  2 |      2 | Getting started |
|  2 | devdojo  |  2 |      2 | Getting started |
|  1 | bobby    |  2 |      2 | Getting started |
|  4 | greisi   |  3 |      3 | SQL is awesome  |
|  3 | tony     |  3 |      3 | SQL is awesome  |
|  2 | devdojo  |  3 |      3 | SQL is awesome  |
|  1 | bobby    |  3 |      3 | SQL is awesome  |
|  4 | greisi   |  4 |      2 | MySQL is up!    |
|  3 | tony     |  4 |      2 | MySQL is up!    |
|  2 | devdojo  |  4 |      2 | MySQL is up!    |
|  1 | bobby    |  4 |      2 | MySQL is up!    |
|  4 | greisi   |  5 |      1 | SQL             |
|  3 | tony     |  5 |      1 | SQL             |
|  2 | devdojo  |  5 |      1 | SQL             |
|  1 | bobby    |  5 |      1 | SQL             |
+----+----------+----+--------+-----------------+
```

As mentioned above, you will highly unlikely run a `CROSS` join for two whole tables in a real-life scenario. If the tables have tens of thousands of rows, an unqualified CROSS JOIN can take minutes to complete.

You would most likely use one of the following with a specific condition.

In MySQL, CROSS JOIN and INNER JOIN are equivalent to JOIN.

INNER JOIN

The `INNER` join is used to join two tables. However, unlike the `CROSS` join, by convention, it is based on a condition. By using an `INNER` join, you can match the first table to the second one.

As we have a one-to-many relationship, a best practice would be to use a primary key for the posts `id` column and a foreign key for the `user_id`; that way, we can 'link' or relate the users table to the posts table. However, this is beyond the scope of this SQL basics eBook, though I might extend it in the future and add more chapters.

As an example and to make things a bit clearer, let's say that you wanted to get all of your users and the posts associated with each user. The query that we would use will look like this:

```
SELECT *
FROM users
INNER JOIN posts
ON users.id = posts.user_id;
```

Rundown of the query:

- `SELECT * FROM users`: This is a standard select we've covered many times in the previous chapters.
- `INNER JOIN posts`: Then, we specify the second table and which table we want to join the result set.
- `ON users.id = posts.user_id`: Finally, we specify how we want the data in these two tables to be merged. The `user.id` is the `id` column of the `user` table, which is also the primary ID, and `posts.user_id` is the foreign key in the email address table referring to the ID column in the users table.

The output will be the following, associating each user with their post based on the `user_id` column:

```
+----+----------+----+---------+-----------------+
| id | username | id | user_id | title           |
+----+----------+----+---------+-----------------+
|  1 | bobby    |  1 |       1 | Hello World!    |
|  2 | devdojo  |  2 |       2 | Getting started |
|  3 | tony     |  3 |       3 | SQL is awesome  |
|  2 | devdojo  |  4 |       2 | MySQL is up!    |
|  1 | bobby    |  5 |       1 | SQL             |
+----+----------+----+---------+-----------------+
```

Note that the INNER JOIN could (in MySQL) equivalently be written merely as JOIN, but that can vary for other SQL dialects:

```
SELECT *
FROM users
JOIN posts
ON users.id = posts.user_id;
```

The main things that you need to keep in mind here are the `INNER JOIN` and `ON` clauses.

With the inner join, the `NULL` values are discarded. For example, if you have a user who does not have a post associated with it, the user with NULL posts will not be displayed when running the above `INNER` join query.

To get the null values as well, you would need to use an outer join.

Types of INNER JOIN

1. **Theta Join (θ)** :- Theta join combines rows from different tables

provided they satisfy the theta condition. The join condition is denoted by the symbol θ.
Here the comparison operators (≤, ≥, □, □, =, ¬) come into picture.
Notation :- $R_1 \bowtie_\theta R_2$.

For example, suppose we want to buy a mobile and a laptop, based on our budget we have thought of buying both such that mobile price should be less than that of laptop.

```
SELECT mobile.model, laptop.model FROM mobile, laptop
WHERE mobile.price < laptop.price;
```

2. **Equijoin** :- When Theta join uses only equality (=) comparison operator, it is said to be equijoin.
   For example, suppose we want to buy a mobile and a laptop, based on our budget we have thought of buying both of the same prices.

   ```
   SELECT mobile.model, laptop.model FROM mobile, laptop
   WHERE mobile.price = laptop.price;
   ```

3. **Natural Join (⋈)** :- Natural join does not use any comparison operator. It does not concatenate the way a Cartesian product does.
   We can perform a Natural Join only if at least one standard column exists between two tables. In addition, the column must have the same name and domain.

   ```
   SELECT * FROM mobile NATURAL JOIN laptop;
   ```

LEFT JOIN

Using the `LEFT OUTER` join, you would get all rows from the first table that you've specified, and if there are no associated records within the second table, you will get a `NULL` value.

In our case, we have a user called `graisi`, which is not associated with a specific post. As you can see from the output from the previous query, the `graisi` user was not present there. To show that user, even though it does not have an associated post with it, you could use a `LEFT OUTER` join:

```
SELECT *
FROM users
LEFT JOIN posts
ON users.id = posts.user_id;
```

The output will look like this:

```
+----+----------+------+---------+-----------------+
| id | username | id   | user_id | title           |
+----+----------+------+---------+-----------------+
|  1 | bobby    |    1 |       1 | Hello World!    |
|  2 | devdojo  |    2 |       2 | Getting started |
|  3 | tony     |    3 |       3 | SQL is awesome  |
|  2 | devdojo  |    4 |       2 | MySQL is up!    |
|  1 | bobby    |    5 |       1 | SQL             |
|  4 | greisi   | NULL |    NULL | NULL            |
+----+----------+------+---------+-----------------+
```

RIGHT JOIN

The `RIGHT OUTER` join is the exact opposite of the `LEFT OUTER` join. It will display all of the rows from the second table and give you a `NULL` value in case that it does not match with an entry from the first table.

Let's create a post that does not have a matching user id:

```
INSERT INTO posts
  ( user_id, title )
VALUES
  ('123', 'No user post!');
```

We specify `123` as the user ID, but we don't have such a user in our `users` table.

Now, if you were to run the `LEFT` outer join, you would not see the post as it has a null value for the corresponding `users` table.

But if you were to run a `RIGHT` outer join, you would see the post but not the `greisi` user as it does not have any posts:

```
SELECT *
FROM users
RIGHT JOIN posts
ON users.id = posts.user_id;
```

Output:

```
+------+----------+----+---------+-----------------+
| id   | username | id | user_id | title           |
+------+----------+----+---------+-----------------+
|    1 | bobby    |  1 |       1 | Hello World!    |
|    2 | devdojo  |  2 |       2 | Getting started |
|    3 | tony     |  3 |       3 | SQL is awesome  |
|    2 | devdojo  |  4 |       2 | MySQL is up!    |
|    1 | bobby    |  5 |       1 | SQL             |
| NULL | NULL     |  6 |     123 | No user post!   |
+------+----------+----+---------+-----------------+
```

Joins can also be limited with WHERE conditions. For instance, in the preceding example, if we wanted to join the tables and then restrict to only username `bobby`.

```
SELECT *
FROM users
RIGHT JOIN posts
ON users.id = posts.user_id
WHERE username = 'bobby';
```

Output:

```
+------+----------+----+---------+-----------------+
| id   | username | id | user_id | title           |
+------+----------+----+---------+-----------------+
|    1 | bobby    |  1 |       1 | Hello World!    |
|    1 | bobby    |  5 |       1 | SQL             |
+------+----------+----+---------+-----------------+
```

The Impact of Conditions in JOIN vs. WHERE Clauses

The placement of conditions within a SQL query, specifically in the `JOIN` vs. the `WHERE` clause, can yield different results.

Take a look at the following example, which retrieves `POSTS` containing the word "SQL" along with their associated user data:

```
SELECT users.*, posts.*
FROM users
LEFT JOIN posts
ON posts.user_id = users.id
WHERE posts.title LIKE '%SQL%';
```

Output:

```
+--+--------+--+-------+-------------------------------+
|id|username|id|user_id|title                          |
+--+--------+--+-------+-------------------------------+
|2 |devdojo |2 |2      |Getting started with SQL       |
|3 |tony    |3 |3      |SQL is awesome                 |
|2 |devdojo |4 |2      |MySQL is up!                   |
|1 |bobby   |5 |1      |SQL - structured query language|
+--+--------+--+-------+-------------------------------+
```

However, by shifting the condition to the `JOIN` clause, all users are displayed, but only posts with titles containing "SQL" are included:

```
SELECT users.*, posts.*
FROM users
LEFT JOIN posts
ON posts.user_id = users.id
   AND posts.title LIKE '%SQL%';
```

Output:

```
+--+--------+----+-------+------------------------------+
|id|username|id  |user_id|title                         |
+--+--------+----+-------+------------------------------+
|1 |bobby   |5   |1      |SQL - structured query language|
|2 |devdojo |4   |2      |MySQL is up!                  |
|2 |devdojo |2   |2      |Getting started with SQL      |
|3 |tony    |3   |3      |SQL is awesome                |
|4 |greisi  |null|null   |null                          |
+--+--------+----+-------+------------------------------+
```

Equivalence of RIGHT and LEFT JOINs

The `RIGHT JOIN` and `LEFT JOIN` operations in SQL are fundamentally equivalent. They can be interchanged by simply swapping the tables involved. Here's an illustration:

The following `LEFT JOIN`:

```
SELECT users.*, posts.*
FROM posts
LEFT JOIN users
ON posts.user_id = users.id;
```

Can be equivalently written using `RIGHT JOIN` as:

```
SELECT users.*, posts.*
FROM users
RIGHT JOIN posts
ON posts.user_id = users.id;
```

Conclusion

Joins are fundamental to using SQL with data. The whole concept of joins might be very confusing initially but would make a lot of sense once you get used to it.

The best way to wrap your head around it is to write some queries, play around with each type of `JOIN`, and see how the result set changes.

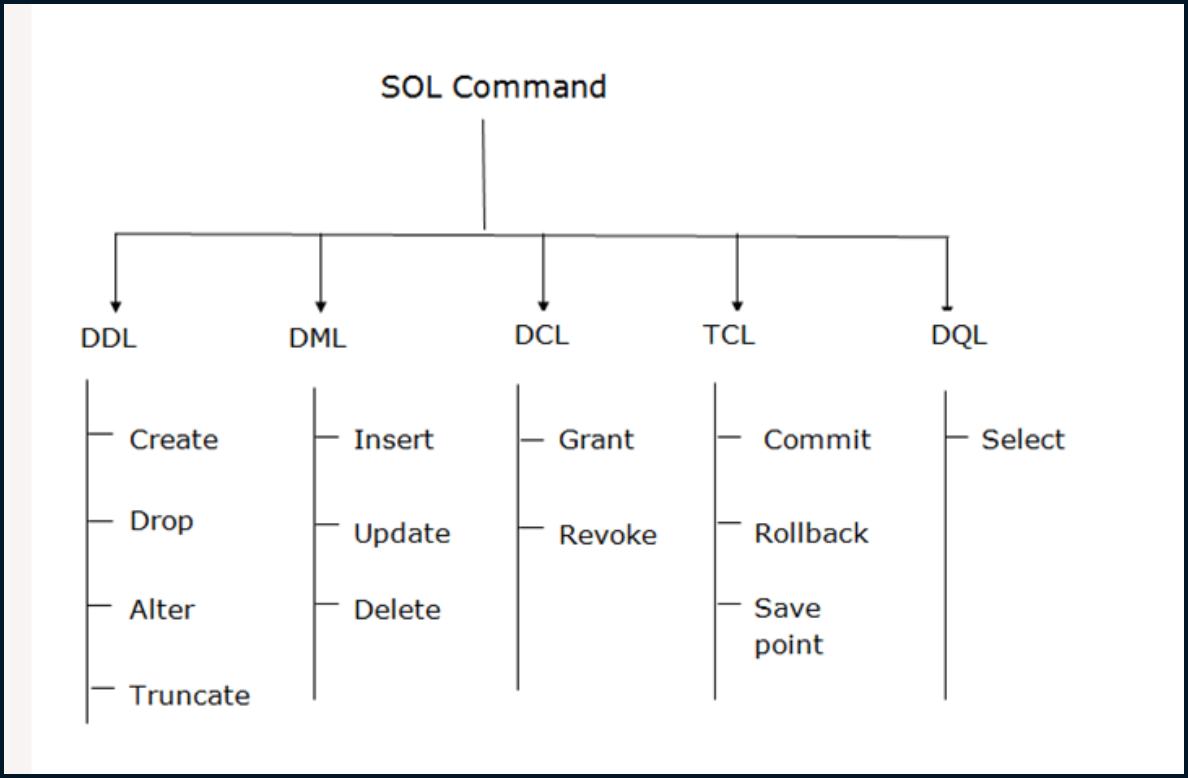
For more information, you could take a look at the official documentation here.

SQL | DDL, DQL, DML, DCL and TCL Commands

Structured Query Language(SQL), as we all know, is the database language by which we can perform certain operations on the existing database. Also, we can use this language to create a database. SQL uses specific commands like Create, Drop, Insert, etc., to carry out the required tasks.

1. **DDL** – Data Definition Language
2. **DQL** – Data Query Language
3. **DML** – Data Manipulation Language
4. **DCL** – Data Control Language

Though many resources claim there to be another category of SQL clauses TCL – Transaction Control Language, so we will see in detail about TCL as well.



DDL (Data Definition Language):

DDL or Data Definition Language consists of the SQL commands used to define the database schema. It simply deals with descriptions of the database schema and is used to create and modify the structure of database objects in the database. These commands usually are not used by a general user, who should be accessing the database via an application.

List of DDL commands:

- `CREATE`: This command is used to create the database or its objects (like table, index, function, views, store procedure, and triggers).

```
CREATE TABLE Persons (
    PersonID int,
    LastName varchar(255),
    FirstName varchar(255),
    Address varchar(255),
    City varchar(255)
);
```

- DROP: This command is used to delete objects from the database.

```
DROP TABLE table_name;
```

- ALTER: This is used to alter the structure of the database.

```
ALTER TABLE Persons
ADD Age int;
```

- TRUNCATE: This is used to remove all records from a table, including all spaces allocated for the records.

```
TRUNCATE TABLE Persons;
```

- COMMENT: This is used to add comments to the data dictionary.

```
--SELECT * FROM Customers;
SELECT * FROM Persons;
```

- RENAME: This is used to rename an object existing in the database.

```
ALTER TABLE Persons
RENAME COLUMN Age TO Year;
```

DQL (Data Query Language):

DQL statements are used for performing queries on the data within schema objects. The purpose of the DQL Command is to get some schema relation based on the query passed to it. We can define DQL as follows. It is a component of the SQL statement that allows getting data from the database and imposing order upon it. It includes the SELECT statement. This command allows getting the data out of the database to perform operations with it. When a SELECT is fired against a table(s), the result is compiled into a different temporary table, which is displayed or perhaps received by the program, i.e. a front-end.

List of DQL:

SELECT: It is used to retrieve data from the database.

```
SELECT * FROM table_name;
```

```
+--------+--------------+------------+--------+---------+
| emp_id | emp_name     | hire_date  | salary | dept_id |
+--------+--------------+------------+--------+---------+
|      1 | Ethan Hunt   | 2001-05-01 |   5000 |       4 |
|      2 | Tony Montana | 2002-07-15 |   6500 |       1 |
|      3 | Sarah Connor | 2005-10-18 |   8000 |       5 |
|      4 | Rick Deckard | 2007-01-03 |   7200 |       3 |
|      5 | Martin Blank | 2008-06-24 |   5600 |    NULL |
+--------+--------------+------------+--------+---------+
```

The SQL commands that deal with the manipulation of data present in the database belong to DML or Data Manipulation Language, including

most of the SQL statements. It is the component of the SQL statement that controls access to data and the database. DCL statements are grouped with DML statements.

List of DML commands:

- `INSERT` : It is used to insert data into a table.

```
INSERT INTO Customers
  (CustomerName, ContactName, Address, City, PostalCode,
Country)
VALUES
  ('Cardinal', 'Tom B. Erichsen', 'Skagen 21', 'Stavanger',
'4006', 'Norway');
```

- `UPDATE`: It is used to update existing data within a table.

```
UPDATE Customers
SET ContactName='Alfred Schmidt', City='Frankfurt'
WHERE CustomerID = 1;
```

- `DELETE` : It is used to delete records from a database table.

```
 DELETE FROM Customers WHERE CustomerName='Alfreds
Futterkiste';
```

- `LOCK`: Table control concurrency.

```
LOCK TABLES table_name [READ | WRITE]
---------------------------------------
UNLOCK TABLES;
```

- `CALL`: Call a PL/SQL or JAVA subprogram.

```
CREATE PROCEDURE procedure_name
AS sql_statement
GO;
```

Execute a Stored Procedure

```
EXEC procedure_name;
```

- `EXPLAIN PLAN`: It describes the access path to data.

DCL (Data Control Language):

DCL includes commands such as GRANT and REVOKE, which mainly deal with the database system's rights, permissions, and other controls.

List of DCL commands:

- `GRANT`: This command gives users access privileges to the database.
- `REVOKE`: This command withdraws the user’s access privileges given by using the GRANT command.

Though many resources claim there to be another category of SQL clauses TCL – Transaction Control Language, we will see in detail about TCL. TCL commands deal with the transaction within the database.

List of TCL commands:

- `COMMIT`: Commits a Transaction.
- `ROLLBACK`: Rollbacks a transaction in case of any error occurs.
- `SAVEPOINT`:Sets a savepoint within a transaction.
- `SET TRANSACTION`: Specify characteristics for the transaction.

SQL Sub Queries

A subquery is a SQL query nested inside a larger query.

- A subquery may occur in
  - A SELECT clause
  - A FROM clause
  - A WHERE clause
- The subquery can be nested inside a SELECT, INSERT, UPDATE, or DELETE statement or inside another subquery.
- A subquery is usually added within the WHERE Clause of another SQL SELECT statement.
- The inner query executes first before its parent query so that the results of an inner query can be passed to the outer query.

You can use a subquery in a SELECT, INSERT, DELETE, or UPDATE statement to perform the following tasks:

- Compare an expression to the result of the query.
- Determine if an expression is included in the results of the query.
- Check whether the query selects any rows.

Subqueries with the `SELECT` Statement:

Consider the CUSTOMERS table having the following records

```
+----+----------+-----+-----------+----------+
| ID | NAME     | AGE | ADDRESS   | SALARY   |
+----+----------+-----+-----------+----------+
|  1 | Ramesh   |  35 | Ahmedabad |  2000.00 |
|  2 | Khilan   |  25 | Delhi     |  1500.00 |
|  3 | Kaushik  |  23 | Kota      |  2000.00 |
|  4 | Chaitali |  25 | Mumbai    |  6500.00 |
|  5 | Hardik   |  27 | Bhopal    |  8500.00 |
|  6 | Komal    |  22 | MP        |  4500.00 |
|  7 | Muffy    |  24 | Indore    | 10000.00 |
+----+----------+-----+-----------+----------+
```

Now, let us check the following subquery with a SELECT statement.

Example:

```
SELECT *
FROM CUSTOMERS
WHERE ID IN (
    SELECT ID
    FROM CUSTOMERS
    WHERE SALARY > 4500
);
```

This would produce the following result.

```
+----+----------+-----+---------+----------+
| ID | NAME     | AGE | ADDRESS | SALARY   |
+----+----------+-----+---------+----------+
|  4 | Chaitali |  25 | Mumbai  |  6500.00 |
|  5 | Hardik   |  27 | Bhopal  |  8500.00 |
|  7 | Muffy    |  24 | Indore  | 10000.00 |
+----+----------+-----+---------+----------+
```

Subqueries with the `UPDATE` Statement:

The subquery can be used in conjunction with the UPDATE statement. Either single or multiple columns in a table can be updated when using a subquery with the UPDATE statement.

Example:

Assuming, we have CUSTOMERS_BKP table available which is backup of CUSTOMERS table. The following example updates SALARY by 0.25 times in the CUSTOMERS table for all the customers whose AGE is greater than or equal to 27.

```
UPDATE CUSTOMERS
SET SALARY = SALARY * 0.25
WHERE AGE IN (
    SELECT AGE
    FROM CUSTOMERS_BKP
    WHERE AGE >= 27
);
```

This would impact two rows and finally CUSTOMERS table would have the following records.

```
+----+----------+-----+-----------+----------+
| ID | NAME     | AGE | ADDRESS   | SALARY   |
+----+----------+-----+-----------+----------+
|  1 | Ramesh   |  35 | Ahmedabad |   125.00 |
|  2 | Khilan   |  25 | Delhi     |  1500.00 |
|  3 | Kaushik  |  23 | Kota      |  2000.00 |
|  4 | Chaitali |  25 | Mumbai    |  6500.00 |
|  5 | Hardik   |  27 | Bhopal    |  2125.00 |
|  6 | Komal    |  22 | MP        |  4500.00 |
|  7 | Muffy    |  24 | Indore    | 10000.00 |
+----+----------+-----+-----------+----------+
```

Subqueries with the `DELETE` Statement:

The subquery can be used in conjunction with the DELETE statement like with any other statements mentioned above.

Example:

Assuming, we have a CUSTOMERS_BKP table available which is a backup of the CUSTOMERS table. The following example deletes the records from the CUSTOMERS table for all the customers whose AGE is greater than or equal to 27.

```
DELETE FROM CUSTOMERS
WHERE AGE IN (
    SELECT AGE
    FROM CUSTOMERS_BKP
    WHERE AGE >= 27
);
```

This would impact two rows and finally the CUSTOMERS table would have the following records.

```
+----+----------+-----+---------+----------+
| ID | NAME     | AGE | ADDRESS | SALARY   |
+----+----------+-----+---------+----------+
|  2 | Khilan   |  25 | Delhi   |  1500.00 |
|  3 | Kaushik  |  23 | Kota    |  2000.00 |
|  4 | Chaitali |  25 | Mumbai  |  6500.00 |
|  6 | Komal    |  22 | MP      |  4500.00 |
|  7 | Muffy    |  24 | Indore  | 10000.00 |
+----+----------+-----+---------+----------+
```

SQL - UNIONS CLAUSE

The SQL UNION clause/operator is used to combine the results of two or more SELECT statements without returning any duplicate rows.

- While using this UNION clause, each SELECT statement must have:
  - The same number of columns selected
  - The same number of column expressions
  - The same data type and
  - Have them in the same order

But they need not have to be in the same length.

Example

Consider the following two tables.

Table 1 − customers table is as follows:

```
+----+----------+-----+-----------+----------+
| id | name     | age | address   | salary   |
+----+----------+-----+-----------+----------+
|  1 | Ramesh   |  32 | Ahmedabad |  2000.00 |
|  2 | Khilan   |  25 | Delhi     |  1500.00 |
|  3 | kaushik  |  23 | Kota      |  2000.00 |
|  4 | Chaitali |  25 | Mumbai    |  6500.00 |
|  5 | Hardik   |  27 | Bhopal    |  8500.00 |
|  6 | Komal    |  22 | MP        |  4500.00 |
|  7 | Muffy    |  24 | Indore    | 10000.00 |
+----+----------+-----+-----------+----------+
```

Table 2 − orders table is as follows:

```
+-----+---------------------+-------------+--------+
| oid | date                | customer_id | amount |
+-----+---------------------+-------------+--------+
| 102 | 2009-10-08 00:00:00 |           3 |   3000 |
| 100 | 2009-10-08 00:00:00 |           3 |   1500 |
| 101 | 2009-11-20 00:00:00 |           2 |   1560 |
| 103 | 2008-05-20 00:00:00 |           4 |   2060 |
+-----+---------------------+-------------+--------+
```

Now, let us join these two tables in our SELECT statement as follows:

```
SELECT id, name, amount, date
   FROM customer
   LEFT JOIN orders
   ON customers.id = orders.customer_id
UNION
   SELECT id, name, amount, date
   FROM customer
   RIGHT JOIN orders
   ON customers.id = orders.customer_id
```

This would produce the following result:

The UNION ALL Clause

The UNION ALL operator is used to combine the results of two SELECT statements including duplicate rows.

The same rules that apply to the UNION clause will apply to the UNION ALL operator.

Example - Consider the following two tables:

- Table 1 − customers table is as follows:

```
+----+----------+-----+-----------+----------+
| id | name     | age | address   | salary   |
+----+----------+-----+-----------+----------+
|  1 | Ramesh   |  32 | Ahmedabad |  2000.00 |
|  2 | Khilan   |  25 | Delhi     |  1500.00 |
|  3 | kaushik  |  23 | Kota      |  2000.00 |
|  4 | Chaitali |  25 | Mumbai    |  6500.00 |
|  5 | Hardik   |  27 | Bhopal    |  8500.00 |
|  6 | Komal    |  22 | MP        |  4500.00 |
|  7 | Muffy    |  24 | Indore    | 10000.00 |
+----+----------+-----+-----------+----------+
```

- Table 2 − orders table is as follows:

```
+-----+---------------------+-------------+--------+
| oid | date                | customer_id | amount |
+-----+---------------------+-------------+--------+
| 102 | 2009-10-08 00:00:00 |           3 |   3000 |
| 100 | 2009-10-08 00:00:00 |           3 |   1500 |
| 101 | 2009-11-20 00:00:00 |           2 |   1560 |
| 103 | 2008-05-20 00:00:00 |           4 |   2060 |
+-----+---------------------+-------------+--------+
```

Now, let us join these two tables in our SELECT statement as follows :

```
SELECT id, name, amount, date
   FROM customers
   LEFT JOIN orders
   ON customers.id = order.customer_id
UNION ALL
   SELECT id, name, amount, date
   FROM customers
   RIGHT JOIN orders
   ON customers.id = orders.customer_id;
```

This would produce the following result:

```
+------+----------+--------+---------------------+
| id   | name     | amount | date                |
+------+----------+--------+---------------------+
|    1 | Ramesh   |   NULL | NULL                |
|    2 | Khilan   |   1560 | 2009-11-20 00:00:00 |
|    3 | kaushik  |   3000 | 2009-10-08 00:00:00 |
|    3 | kaushik  |   1500 | 2009-10-08 00:00:00 |
|    4 | Chaitali |   2060 | 2008-05-20 00:00:00 |
|    5 | Hardik   |   NULL | NULL                |
|    6 | Komal    |   NULL | NULL                |
|    7 | Muffy    |   NULL | NULL                |
|    3 | kaushik  |   3000 | 2009-10-08 00:00:00 |
|    3 | kaushik  |   1500 | 2009-10-08 00:00:00 |
|    2 | Khilan   |   1560 | 2009-11-20 00:00:00 |
|    4 | Chaitali |   2060 | 2008-05-20 00:00:00 |
+------+----------+--------+---------------------+
```

Note : **There are two other clauses (i.e., operators), which are like the UNION clause.**

Relational Keys- Keys in a Relational Database

A database must be able to inhibit inconsistency occurring due to incorrect data. It must have certain identified attributes in relations to uniquely distinguish the tuples. No two tuples in a relation should have same value for all attributes since it will lead to duplicity of data. duplicity of data leads to inconsistency . Relational database systems have the concept of Relational Keys to distinguish between different records.

Types of Relational Keys

Super Keys

A relation’s tuples can be uniquely identified by various combinations of attributes. Super Keys is defined as a set of one attribute or combinations of two or more attributes that help in distinguishing between tuples in a relation.

For example, the Customer ID attribute of the relation Customer is unique for all customers. The Customer ID can be used to identify each customer tuple in the relation. Customer ID is a Super Key for relation Customer.

Customer Name attribute of Customer cannot be considered as Super Key because many customers for the organization can have same name. However when combined with Customer ID it becomes a Super Key {CustomerID, CustomerName}. It means that Super Key can have additional attributes. Consider any key K which is identified as a super key. Any superset of key K is also a super key. For example the possible Super Keys for Customer Relation are

- [CustomerID, CustomerName, Customer Address]
- [CustomerID, CustomerName, Customer Contact Number]
- [CustomerID, Customer Contact Number]

Candidate Keys

If we take a key from the set of super keys for which we don’t have any proper subset defined as a superkey, it is called a candidate key. In other words the minimal attribute super keys are termed as candidate keys.

If we can identify some distinct sets of attributes which identify the tuples uniquely they fall in the category of candidate keys. For example the possible Candidate Keys for Customer Relation are

- [CustomerID]
- [CustomerName, Customer Address]
- [CustomerName, Customer Contact Number]
- [Customer Address, Customer Contact Number]

Primary Key

Out of all possible candidate keys only one is chosen by the database designer as the key to identify the records in a relation in a database. This selected candidate key is called the Primary Key. It is the property of the relation and not of tuples. The primary key attribute(s) does not allow any duplicate values. It also inhibits leaving the primary key attribute without any value (NOT NULL).

A relation can have only one primary key.

In the Customer Database example {Customer ID} is the attribute taken as the primary key of customer relation. While picking up a candidate key as primary key the designer should ensure that it is an attribute or group of attributes that do not change or may change extremely rarely.

Alternate Keys

After selecting one key among candidate keys as primary key, the rest of candidate keys are called the alternate keys. In the customer Database these candidate keys are the alternate keys.

- [CustomerName, Customer Address]

- [CustomerName, Customer Contact Number]
- [Customer Address, Customer Contact Number]

Foreign Key

A foreign key is used to reference values from one relation into another relation. This is possible when the attribute or combination of attributes is primary key in the referenced relation. The relation in which the primary key of a relation is referenced is called the referencing table. The foreign key constraint implements the referential integrity in a database. The referencing relation attribute can have only those values which exist in the primary key attribute(s) of the referenced relation

A relation can have multiple foreign key

For example in the customer database the orders' relation (referencing relation) has the structure (Order ID, Customer ID, Order Date, Order Status, Total Billing Amount). The attribute Customer ID is the foreign key referencing Customer ID from customer relation (referenced relation). It means that orders can be placed only for the customers whose customer details are already available in the customer relation.

Logical Operator Keywords

Here are the most important Logical Operators summarized in a table.

Logical Operators can be used for conditions as they show a result in form of a `boolean` (True/False) or Unknown. So, e.g. if an exact value is `True` for a value, a Logical Operator can proof that it's True.

Logical Operator	Explanation
ALL	If all comparisons are True: return True
ANY	If any comparison is True: return True
AND	If both expressions are True: return True
EXISTS	If a subquery contains rows: return True
IN	If compared value is equal to at least one value: return True
BETWEEN	If there are values in given range: return True
NOT	Reverses the value of any boolean
OR	If either expression is True: return True

HAVING Clause

Unlike where clause which imposes conditions on columns `Having` clause enables you to specify conditions that filter which group results appear in the results.

Syntax

```
SELECT column_name(s)
FROM table_name
WHERE condition
GROUP BY column_name(s)
HAVING condition
ORDER BY column_name(s);
```

Description

- Used with `aggregate functions`
- Must follow `GROUP BY` clause in the query

Aggregate Functions

- SQL aggregation is the task of collecting a set of values to return a single value.
- An aggregate function is a function where the values of multiple rows are grouped together as input on certain criteria to form a single value of more significant meaning.

Aggregate Functions Examples

Suppose this are the table given to us

Students table

rollno	name	class
1	Sanskriti	TE
1	Shree	BE
2	Harry	TE
3	John	TE
3	Shivani	TE

purchase table

item	price	customer_name
Pen	10	Sanskriti
Bag	1000	Sanskriti
Vegetables	500	Sanskriti
Shoes	5000	Sanskriti
Water Bottle	800	XYZ
Mouse	120	ABC
Sun Glasses	1350	ABC

AVG function

Calculates `average` of the given column of values

```
SELECT AVG(price) AS Avg_Purchase, customer_name
FROM purchase
GROUP BY customer_name;
```

Avg_Purchase	customer_name
1627.5000	Sanskriti

SUM function

Calculates `sum` of values of given column.

```
SELECT SUM(price) AS Total_Bill, customer_name
FROM purchase
GROUP BY customer_name;
```

Total_Bill	customer_name
6510	Sanskriti

COUNT function

Gives `count` of entries/ values in given column.

```
SELECT COUNT(item) AS Total_Items, customer_name
FROM purchase
GROUP BY customer_name;
```

Total_Items	customer_name
4	Sanskriti

MAX function

Return `maximum` value from the number of values in the column.

```
SELECT MAX(price) AS Highest_Purchase, customer_name
FROM purchase
GROUP BY customer_name;
```

Highest_Purchase	customer_name
5000	Sanskriti

MIN function

Return `minimum` value from the number of values in the column.

```
SELECT MIN(price) AS Lowest_Purchase, customer_name
FROM purchase
GROUP BY customer_name;
```

Lowest_Purchase	customer_name
10	Sanskriti

Having clause Examples

Example 1

```
SELECT COUNT(class) AS strength, class
FROM Students
GROUP BY class
HAVING COUNT(class) > 2;
```

Above query gives number of students in a class `having` number of students > 2

strength	class
4	TE

Example 2

```
SELECT customer_name, MIN(price) AS MIN_PURCHASE
FROM purchase
GROUP BY customer_name
HAVING MIN(price) > 10;
```

Above query finds `minimum` price which is > 10

customer_name	MIN_PURCHASE
XYZ	800
ABC	120

Example 3

```
SELECT customer_name, AVG(price) AS Average_Purchase
FROM purchase
GROUP BY customer_name
HAVING AVG(price) > 550
ORDER BY customer_name DESC;
```

Above query calculates `average` of price and prints customer name and average price which is greater than 550 with descending `order` of customer names.

customer_name	Average_Purchase
XYZ	800.0000
Sanskriti	1627.5000
ABC	735.0000

Example 4

```
SELECT customer_name, SUM(price) AS Total_Purchase
FROM purchase
WHERE customer_name
LIKE "S%"
GROUP BY customer_name
HAVING SUM(price) > 1000;
```

Calculates `SUM` of price and returns customer name and sum > 1000.

customer_name	Total_Purchase
Sanskriti	6510

Essential MySQL Functions

MySQL has many built-in functions. We will covering some important most used built-in functions; for a complete list refer to the online MySQL Reference Manual (http://dev.mysql.com/doc/).

NOTE: As of now we will be going through only function and their output, as they would be self explanatory.

Numeric Functions

```
SELECT ROUND(5.73)
```

6

```
SELECT ROUND(5.73, 1)
```

5.7

```
SELECT TRUNCATE(5.7582, 2)
```

5.75

```
SELECT CEILING(5.2)
```

6

```
SELECT FLOOR(5.7)
```

5

```
SELECT ABS(-5.2)
```

5.2

```
SELECT RAND() -- Generates a random floating point number b/w
0 & 1
```

STRING Functions

```
SELECT LENGTH('sky')
```

3

```
SELECT UPPER('sky')
```

SKY

```
SELECT LOWER('sky)
```

sky

```
SELECT LTRIM('    sky')
```

sky

```
SELECT RTRIM('sky    ')
```

sky

```
SELECT TRIM('   sky  ')
```

sky

```
SELECT LEFT('Kindergarten', 4)
```

Kind

```
SELECT RIGHT('Kindergarten', 6)
```

garten

```
SELECT SUBSTRING('Kindergarten', 3, 5)
```

nderg

```
SELECT LOCATE('n','Kindergarten') -- LOCATE returns the first
occurrence of a character or character string, if found,
otherwise it returns 0
```

3

```
SELECT REPLACE('Kindergarten', 'garten', 'garden')
```

Kindergarden

```
SELECT CONCAT('first', 'last')
```

firstlast

DATE Functions

```
SELECT NOW()
```

2021-10-21 19:59:47

```
SELECT CURDATE()
```

2021-10-21

```
SELECT CURTIME()
```

20:01:12

```
SELECT MONTH(NOW())
```

10

```
SELECT YEAR(NOW())
```

2021

```
SELECT HOUR(NOW())
```

13

```
SELECT DAYTIME(NOW())
```

Thursday

Formatting Dates and Times

In MySQL, the default date format is "YYYY-MM-DD", ex: "2025-05-12", MySQL allows developers to format it the way they want. We will discuss some of them.

```
SELECT DATE_FORMAT(NOW(), '%M %D %Y')
```

October 22nd 2021

```
SELECT DATE_FORMAT(NOW(), '%m %d %y')
```

10 22 21

```
SELECT DATE_FORMAT(NOW(), '%m %D %y')
```

10 22nd 21

```
SELECT TIME_FORMAT(NOW(), '%H %i %p')
```

14:11 PM

Calculating Dates and Times

```
SELECT DATE_ADD(NOW(), INTERVAL 1 DAY) --return tomorrows date and time
```

2021-10-23 14:26:17

```
SELECT DATE_ADD(NOW(), INTERVAL -1 YEAR)
```

or

```
SELECT DATE_SUB(NOW(), INTERVAL 1 YEAR)
```

Both the queries will return the same output

2020-10-22 14:29:47

```
SELECT DATEDIFF('2021-09-08 09:00', '2021-07-07 17:00') -- It will return the difference in number of days, time won't be considered
```

63

```
SELECT TIME_TO_SEC('09:00') - TIME_TO_SEC('09:02')
```

-120

Triggers In SQL

A `trigger` is a stored procedure in database which is automatically invoked whenever any special event occurs in the database. The event can be any event including INSERT, UPDATE and DELETE.

For eg: If you want to perfom a task after a record is inserted into the table then we can make use of `triggers`

Syntax for creating triggers

```
create trigger [trigger_name]
[before | after]
{insert | update | delete}
on [table_name]
[for each row | for each column]
[trigger_body]
```

`create trigger [trigger_name]` : Creates or replaces an existing trigger with the trigger_name.

`[before | after]` : Now we can specify when our trigger will get fired. It can be before updating the database or after updating the database.

Generally , `before` triggers are used to validate the data before storing it into the database.

`{insert | update | delete}` : Now, we specify the `DML operation` for which our trigger should get fired .

`on [table_name]` : Here, we specify the name of the table which is

associated with the trigger.

`[for each row]` : This specifies a row-level trigger, i.e., the trigger will be executed for each row being affected.

`[for each column]` : This specifies a column-level trigger, i.e., the trigger will be executed after the specified column is affected.

`[trigger_body]` : Here, we specify the operations to be performed once the trigger is fired.

Show Trigger

If you want to see all the triggers that are present in your database.

```
show triggers in database_name;
```

Drop Trigger

if you no longer want your trigger then you may delete it.

```
drop trigger trigger_name;
```

Example :

Let us consider we have our database named `library`. Consider a scenario where we want a trigger which is fired everytime any particular book is inserted into the `books` table . The `trigger` should add the logs of all the books that are inserted into the `books` table.

We have created two tables :

1. `books` : It will store all the books available in the library
2. bookrecord : It will generate a statement a log for the inserted book

```
Select * from library.books;
```

```
+----------+---------------+
| book_id  | book_name     |
+----------+---------------+
|          |                   |
|          |                   |
+----------+---------------+
```

Here, `book_id` is an auto-incremental field.

```
Select * from library.bookrecord;
```

```
+----------+---------------+-----------+
| SRNO     | bookid        | statement |
+----------+---------------+-----------+
|          |               |           |
|          |               |           |
+----------+---------------+-----------+
```

Here, SRNO is an auto-incremental field.

Now, we will create our trigger on the books table

```
create trigger library.addstatement
after insert
on library.books
for each row
insert into library.bookrecord(bookid,statement) values
(NEW.book_id,concat('New book named ',NEW.book_name,"  added
at ",curdate()));
```

In MySQL, NEW is used to access the currently inserted row. We are inserting the log for the currently inserted book in our database.

Now we will insert a book and wait for the output.

```
insert into library.books(book_name) values ("Harry Potter and
the Goblet of fire");
```

Output for books:

```
+----------+-------------------------------------------+
| book_id  | book_name                                 |
+----------+-------------------------------------------+
|    1     |         Harry Potter and the Goblet of fire
|
|          |
|
+----------+-------------------------------------------+
```

Output for `bookrecord`:

```
+----------+---------------+-----------------------------------
-----------------------------------------------+
| SRNO     | bookid        | statement
|
+----------+---------------+-----------------------------------
-----------------------------------------------+
|   1      |      1        |     New book named Harry Potter
and the Goblet of fire  added at 2021-10-22      |
|          |               |
|
+----------+---------------+-----------------------------------
-----------------------------------------------+
```

See. it worked!!

Conclusion:

Here, you learnt what are triggers and how you create them. You can create different types of triggers based on your needs and requirements.

Transaction Control Language

- `Transaction Control Language` can be defined as the portion of a database language used for `maintaining consistency` of the database and `managing transactions` in the database.

- A set of `SQL statements` that are `co-related logically and executed on the data stored in the table` is known as a `transaction`.

TCL Commands

- COMMIT Command
- ROLLBACK Command
- SAVEPOINT Command

COMMIT

The main use of `COMMIT` command is to `make the transaction permanent`. If there is a need for any transaction to be done in the database that transaction permanent through commit command.

Syntax

```
COMMIT;
```

ROLLBACK

Using this command, the database can be `restored to the last committed state`. Additionally, it is also used with savepoint command for jumping to a savepoint in a transaction.

Syntax

```
ROLLBACK TO savepoint-name;
```

SAVEPOINT

The main use of the Savepoint command is to save a transaction temporarily. This way users can rollback to the point whenever it is needed.

Syntax

```
SAVEPOINT savepoint-name;
```

Examples

This is purchase table that we are going to use through this tutorial

item	price	customer_name
Pen	10	Sanskriti
Bag	1000	Sanskriti
Vegetables	500	Sanskriti
Shoes	5000	Sanskriti
Water Bottle	800	XYZ
Mouse	120	ABC
Sun Glasses	1350	ABC

```
UPDATE purchase SET price = 20 WHERE item = "Pen";
```

O/P : Query OK, 1 row affected (3.02 sec) (Update the price of Pen set it from 10 to 20)

```
SELECT * FROM purchase;
```

O/P

item	price	customer_name
Pen	20	Sanskriti
Bag	1000	Sanskriti
Vegetables	500	Sanskriti
Shoes	5000	Sanskriti
Water Bottle	800	XYZ
Mouse	120	ABC
Sun Glasses	1350	ABC

```
START TRANSACTION;
```

Start transaction

```
COMMIT;
```

Saved/ Confirmed the transactions till this point

```
ROLLBACK;
```

Lets consider we tried to rollback above transaction

```
SELECT * FROM purchase;
```

O/P:

item	price	customer_name
Pen	20	Sanskriti
Bag	1000	Sanskriti
Vegetables	500	Sanskriti
Shoes	5000	Sanskriti
Water Bottle	800	XYZ
Mouse	120	ABC
Sun Glasses	1350	ABC

As we have committed the transactions the `rollback` will not affect anything

```
SAVEPOINT  sv_update;
```

Create the `savepoint` the transactions above this will not be rollbacked

```
UPDATE purchase SET price = 30 WHERE item = "Pen";
```

O/P : Query OK, 1 row affected (0.57 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
SELECT * FROM purchase;
```

item	price	customer_name
Pen	30	Sanskriti
Bag	1000	Sanskriti
Vegetables	500	Sanskriti
Shoes	5000	Sanskriti
Water Bottle	800	XYZ
Mouse	120	ABC
Sun Glasses	1350	ABC

price of pen is changed to 30 using the `update` command

```
ROLLBACK to sv_update;
```

Now if we `rollback` to the `savepoint` price should be 20 after `rollback` lets see

```
SELECT * FROM purchase;
```

item	price	customer_name
Pen	20	Sanskriti

item	price	customer_name
Bag	1000	Sanskriti
Vegetables	500	Sanskriti
Shoes	5000	Sanskriti
Water Bottle	800	XYZ
Mouse	120	ABC
Sun Glasses	1350	ABC
Torch	850	ABC

As expected we can see `update` query is rollbacked to sv_update.

Conclusion

With this short tutorial we have learnt TCL commands.

Data Control Language

DCL commands are used to grant and take back authority from any database user.

DCL Commands

- GRANT Command
- REVOKE Command

GRANT

GRANT is used to give user access privileges to a database.

Syntax

```
GRANT privilege_name ON objectname TO user;
```

REVOKE

REVOKE remove a privilege from a user. REVOKE helps the owner to cancel previously granted permissions.

Syntax

```
REVOKE privilege_name ON objectname FROM user;
```

DCL Examples

```
SELECT * FROM purchase;
```

Output:

```
| item         | price | customer_name |
|--------------|-------|---------------|
| Pen          |    20 | Sanskriti     |
| Bag          |  1000 | Sanskriti     |
| Vegetables   |   500 | Sanskriti     |
| Shoes        |  5000 | Sanskriti     |
| Water Bottle |   800 | XYZ           |
| Mouse        |   120 | ABC           |
| Sun Glasses  |  1350 | ABC           |
| Torch        |   850 | ABC           |
```

- Lets start with GRANT command:

```
  GRANT INSERT ON purchase TO 'Sanskriti'@'localhost';
```

Output:

```
#### O/P Query OK, 0 rows affected (0.31 sec)
```

Description In above command we have granted user Sanskriti priviledge to `Insert` into purchase table.

- Now if I login as Sanskriti and try to run `Select` statement as given below what should happen?

```
SELECT * FROM purchase;
```

Output:

```
#### O/P ERROR 1142 (42000): SELECT command denied to user
'Sanskriti'@'localhost' for table 'purchase'
```

Yup as expected it gives error because we have granted insert operation to Sanskriti.

- So lets try inserting data to purchase table:

```
INSERT INTO purchase values("Laptop", 100000, "Sanskriti");
```

Output:

```
#### O/P Query OK, 1 row affected (0.34 sec)
```

Yes! It works!

- Now I am checking the purchase table from my original account:

```
SELECT * FROM purchase;
```

Output:

```
| item         | price  | customer_name |
|--------------|--------|---------------|
| Pen          |     20 | Sanskriti     |
| Bag          |   1000 | Sanskriti     |
| Vegetables   |    500 | Sanskriti     |
| Shoes        |   5000 | Sanskriti     |
| Water Bottle |    800 | XYZ           |
| Mouse        |    120 | ABC           |
| Sun Glasses  |   1350 | ABC           |
| Torch        |    850 | ABC           |
| Laptop       | 100000 | Sanskriti     |
```

As you can see, the row is inserted.

- Now lets try `Revoke` command:

```
REVOKE INSERT ON purchase FROM 'Sanskriti'@'localhost';
```

Output:

```
#### O/P Query OK, 0 rows affected (0.35 sec)
```

Now we have revoked the insert priviledge from Sanskriti.

- If Sanskriti runs insert statement it should give error:

```
INSERT INTO purchase values("Laptop", 100000, "Sanskriti");
```

Output:

```
#### O/P ERROR 1142 (42000): INSERT command denied to user
'Sanskriti'@'localhost' for table 'purchase'
```

Conclusion

Through this tutorial we have learnt `DCL` commands and their usage.

The MySQL dump command

There are many ways and tools on how to export or backup your MySQL databases. In my opinion, `mysqldump` is a great tool to accomplish this task.

The mysqldump tool can be used to dump a database or a collection of databases for backup or transfer to another database server (not necessarily MariaDB or MySQL). The dump typically contains SQL statements to create the table, populate it, or both.

One of the main benefits of mysqldump is that it is available out of the box on almost all shared hosting servers. So if you are hosting your database on a cPanel server that you don't have root access to, you could still use it to export more extensive databases.

Exporting a Database

To export/backup a database, all you need to do is run the following command:

```
mysqldump -u your_username -p your_database_name >
your_database_name-$(date +%F).sql
```

Note that you need to change the your_database_name with the actual name of your database and the `your_username` part with your existing MySQL username.

Rundown of the arguments:

- `-u`: needs to be followed by your MySQL username
- `-p`: indicates that you would be prompted for your MySQL password
- `>`: indicates that the output of the command should be stored in the .sql file that you specify after that sign

You would create an export of your database by running the above command, which you could later use as a backup or even transfer it to another server.

Exporting all databases

If you have root access to the server, you could use the `--all-databases flag to export all of the databases hosted on the particular MySQL server. The downside of this approach is that this would create one single` .sql` export, which would contain all of the databases.

Let's say that you would like to export each database into a separate .sql file. You could do that with the following script:

```
#!/bin/bash

##
# Get a list of all databases except the system databases that
are not needed
##
DATABASES=$(echo "show databases;" | mysql | grep -Ev
"(Database|information_schema|mysql|performance_schema)")

DATE=$(date +%d%m%Y)
TIME=$(date +%s)
BACKUP_DIR=/home/your_user/backup

##
# Create Backup Directory
##

if [ ! -d ${BACKUP_DIR} ]; then
  mkdir -p ${BACKUP_DIR}
fi

##
# Backup all databases
##

for DB in $DATABASES;
do
    mysqldump --single-transaction --skip-lock-tables $DB |
gzip > ${BACKUP_DIR}/$DATE-$DB.sql.gz
done
```

The script would backup each database and store the .sql dumps in the `/home/your_user/backup` folder. Make sure to adjust the path to your backup folder.

For more information on Bash scripting, check out this opensource eBook here.

Automated backups

You can even set a cronjob to automate the backups above; that way, you would have regular backups of your databases.

To do that, you need to make sure that you have the following content in your `.my.cnf` file. The file should be stored at:

```
/home/your_user/.my.cnf
```

You should make sure that it has secure permissions:

```
chmod 600 /home/your_user/.my.cnf
```

And you should add the following content:

```
[client]
user=your_mysql_user
password=your_mysql_password
```

Once you have your `.my.cnf` file configured, you set up a cronjob to trigger the mysqldump export of your database:

```
0 10,22 * * * /usr/bin/mysqldump -u your_username -p
your_database_name > your_database_name-$(date +%F).sql
```

The above would run at 10 AM and 10 PM every day, so you will have

two daily database backups.

You can even expand the logic and add a compression step so that the .sql dumps do not consume too much webspace.

Conclusion

The `mysqldump` is a great tool to easily and quickly backup your MySQL databases.

For more information, you could take a look at the official documentation here:

- mysqldump

This was initially posted here.

Learn Materialize by running streaming SQL on your nginx logs

In this tutorial, I will show you how Materialize works by using it to run SQL queries on continuously produced nginx logs. By the end of the tutorial, you will have a better idea of what Materialize is, how it's different than other SQL engines, and how to use it.

Prerequisites

For the sake of simplicity, I will use a brand new Ubuntu 21.04 server where I will install nginx, Materialize and `mzcli`, a CLI tool similar to `psql` used to connect to Materialize and execute SQL on it.

If you want to follow along you could spin up a new Ubuntu 21.04 server on your favorite could provider.

If you prefer runing Materialize on a different operating system, you can follow the steps on how to install Materialize here:

- How to install Materialize

What is Materialize

Materialize is a streaming database for real-time analytics.

It is not a substitution for your transactional database, instead it accepts input data from a variety of sources like:

- Messages from streaming sources like Kafka
- Archived data from object stores like S3
- Change feeds from databases like PostgreSQL
- Data in Files: CSV, JSON and even unstructured files like logs *(what we'll be using today.)*

And it maintains the answers to your SQL queries over time, keeping them up-to-date as new data flows in (using *materialized views*), instead of running them against a static snapshot at a point in time.



If you want to learn more about Materialize, make sure to check out their official documentation here:

Materialize Documentation

Installing Materialize

Materialize runs as a single binary called `materialized` *(d for daemon, following Unix conventions)*. Since we're running on Linux, we'll just install Materialize directly. To install it, run the following command:

```
sudo apt install materialized
```

Once it's installed, start Materialize (with sudo so it has access to nginx logs):

```
sudo materialized
```

Now that we have the `materialized` running, we need to open a new terminal to install and run a CLI tool that we use to interact with our Materialize instance!

There are other ways that you could use in order to run Materialize as described here. For a production-ready Materialize instance, I would recommend giving Materialize Cloud a try!

Installing `mzcli`

The `mzcli tool` lets us connect to Materialize similar to how we would use a SQL client to connect to any other database.

Materialize is wire-compatible with PostgreSQL, so if you have `psql` already installed you could use it instead of `mzcli`, but with `mzcli` you get nice syntax highlighting and autocomplete when writing your queries.

To learn the main differences between the two, make sure to check out the official documentation here: Materialize CLI Connections

The easiest way to install `mzcli` is via `pipx`, so first run:

```
apt install pipx
```

and, once `pipx` is installed, install `mzcli` with:

```
pipx install mzcli
```

Now that we have `mzcli` we can connect to `materialized` with:

```
mzcli -U materialize -h localhost -p 6875 materialize
```



For this demo, let's quickly install nginx and use Regex to parse the log and create Materialized Views.

Installing nginx

If you don't already have nginx installed, install it with the following command:

```
sudo apt install nginx
```

Next, let's populate the access log with some entries with a Bash loop:

```
for i in {1..200} ; do curl -s 'localhost/materialize'  >
/dev/null ; echo $i ; done
```

If you have an actual nginx `access.log`, you can skip the step above.

Now we'll have some entries in the `/var/log/nginx/access.log` access log file that we would be able to able to feed into Materialize.

Adding a Materialize Source

By creating a Source you are essentially telling Materialize to connect to some external data source. As described in the introduction, you could connect a wide variety of sources to Materialize.

For the full list of source types make sure to check out the official documentation here:

Materialize source types

Let's start by creating a text file source from our nginx access log.

First, access the Materialize instance with the `mzcli` command:

```
mzcli -U materialize -h localhost -p 6875 materialize
```

Then run the following statement to create the source:

```
CREATE SOURCE nginx_log
FROM FILE '/var/log/nginx/access.log'
WITH (tail = true)
FORMAT REGEX '(?P<ipaddress>[^ ]+) - - \[(?P<time>[^\]]+)\]
"(?P<request>[^ ]+) (?P<url>[^ ]+)[^"]+"
(?P<statuscode>\d{3})';
```

A quick rundown:

- `CREATE SOURCE`: First we specify that we want to create a source
- `FROM FILE`: Then we specify that this source will read from a local file, and we provide the path to that file
- `WITH (tail = true)`: Continually check the file for new content
- `FORMAT REGEX`: as this is an unstructured file we need to specify

regex as the format so that we could get only the specific parts of the log that we need.

Let's quickly review the Regex itself as well.

The Materialize-specific behavior to note here is the `?P<NAME_HERE>` pattern extracts the matched text into a column named `NAME_HERE`.

To make this a bit more clear, a standard entry in your nginx access log file would look like this:

```
123.123.123.119 - - [13/Oct/2021:10:54:22 +0000] "GET /
HTTP/1.1" 200 396 "-" "Mozilla/5.0 zgrab/0.x"
```

- `(?P<ipaddress>[^ ]+)`: With this pattern we match the IP address for each line of the nginx log, e.g. `123.123.123.119`.
- `\[(?P<time>[^\]]+)\]`: the timestamp string from inside square brackets, e.g. `[13/Oct/2021:10:54:22 +0000]`
- `"(?P<request>[^ ]+)`: the type of request like `GET`, `POST` etc.
- `(?P<url>[^ ]+)`: the relative URL, eg. `/favicon.ico`
- `(?P<statuscode>\d{3})`: the three digit HTTP status code.

Once you execute the create source statement, you can confirm the source was created successfully by running the following:

```
mz> SHOW SOURCES;
// Output
+-----------+
| name      |
|-----------|
| nginx_log |
+-----------+
SELECT 1
Time: 0.021s
```

Now that we have our source in place, let's go ahead and create a view!

Creating a Materialized View

You may be familiar with Materialized Views from the world of traditional databases like PostgreSQL, which are essentially cached queries. The unique feature here is the materialized view we are about to create is **automatically kept up-to-date**.

In order to create a materialized view, we will use the following statement:

```
CREATE MATERIALIZED VIEW aggregated_logs AS
  SELECT
    ipaddress,
    request,
    url,
    statuscode::int,
    COUNT(*) as count
  FROM nginx_log GROUP BY 1,2,3,4;
```

The important things to note are:

- Materialize will keep the results of the embedded query in memory, so you'll always get a fast and up-to-date answer
- The results are incrementally updated as new log events arrive

Under the hood, **Materialize compiles your SQL query into a dataflow** and then takes care of all the heavy lifting for you. This is incredibly powerful, as it allows you to process data in real-time using *just* SQL.

A quick rundown of the statement itself:

- First we start with the `CREATE MATERIALIZED VIEW aggregated_logs` which identifies that we want to create a new Materialized view named `aggregated_logs`.
- Then we specify the `SELECT` statement that we are interested in keeping track of over time. In this case we are aggregating the data in our log file by `ipaddress`, `request`, `url` and `statuscode`, and we are counting the total instances of each combo with a `COUNT(*)`

When creating a Materialized View, it could be based on multiple sources like a stream from Kafka, a raw data file that you have on an S3 bucket, or your PostgreSQL database. This single statement will give you the power to analyze your data in real-time.

We specified a simple `SELECT` that we want the view to be based on but this could include complex operations like `JOIN`s, however for the sake of this tutorial we are keeping things simple.

For more information about Materialized Views check out the official documentation here:

Creating Materialized views

Now you could use this new view and interact with the data from the nginx log with pure SQL!

Reading from the view

If we do a SELECT on this Materialized view, we get a nice aggregated summary of stats:

```
SELECT * FROM aggregated_logs;

   ipaddress    | request |           url            |
statuscode | count
----------------+---------+--------------------------+--------
----+-------
 127.0.0.1      | GET     | /materialize             |
404 |   200
```

As more requests come in to the nginx server, the aggregated stats in the view are kept up-to-date.

We could also write queries that do further aggregation and filtering on top of the materialized view, for example, counting requests by route only:

```
SELECT url, SUM(count) as total FROM aggregated_logs GROUP BY
1 ORDER BY 2 DESC;
```

If we were re-run the query over and over again, we could see the numbers change instantly as soon as we get new data in the log as Materialize processes each line of the log and keeps listening for new lines:

```
+--------------------------+-------+
| url                      | total |
|--------------------------+-------|
| /materialize/demo-page-2 | 1255  |
| /materialize/demo-page   | 1957  |
| /materialize             | 400   |
+--------------------------+-------+
```

As another example, let's use `psql` together with the `watch` command to see this in action.

If you don't have `psql` already isntalled you can install it with the following command:

```
sudo apt install postgresql-client
```

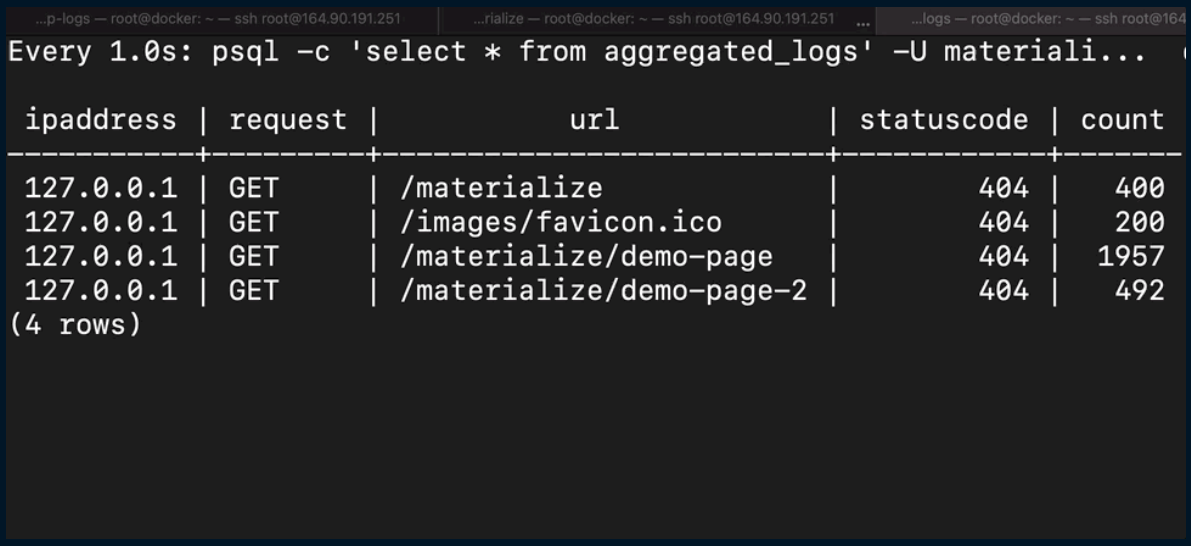
After that, let's run the `SELECT * FROM aggregated_logs` statement every second using the `watch` command:

```
watch -n1 "psql -c 'select * from aggregated_logs' -U
materialize -h localhost -p 6875 materialize"
```

In **another terminal window**, you could run another `for` loop to generate some new nginx logs and see how the results change:

```
for i in {1..2000} ; do curl -s 'localhost/materialize/demo-
page-2' > /dev/null ; echo $i ; done
```

The output of the `watch` command would look like this:



Feel free to experiment with more complex queries and analyze your nginx access log for suspicious activity using pure SQL and keep track of the results in real-time!

Conclusion

By now, hopefully you have a hands-on understanding of how incrementally maintained materialized views work in Materialize. In case that you like the project, make sure to star it on GitHub:

https://github.com/MaterializeInc/materialize

Source

Conclusion

Congratulations! You have just completed the SQL basics guide!

If you found this helpful, be sure to star the project on GitHub!

If you have any suggestions for improvements, make sure to contribute pull requests or open issues.

In this introduction to SQL eBook, we just covered the basics, but you still have enough under your belt to start working with any relational database out there!

As a next step, try to create a database server, import some demo data, and play around with all of the statements that you've learned so far. You can also take a look at this website here to help you build your tables and relations and visualize them:

https://dbdiagram.io/

In case that this eBook inspired you to contribute to some fantastic open-source project, make sure to tweet about it and tag @bobbyiliev_ so that we could check it out!

Congrats again on completing this eBook!

Other eBooks

Some other opensource eBooks that you might find helpful are:

- Introduction to Git and GitHub
- Introduction to Bash Scripting

SQL Injection — A Beginner's Understanding

Purpose: This guide provides beginners with a clear, safe, and practical overview of what SQL injection (SQLi) is, how it happens at a conceptual level, and — most importantly — how to prevent and detect it. It focuses on defensive, responsible practices without providing exploit instructions.

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1. What is SQL Injection?
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6. Top prevention techniques (detailed)
7. Detection, logging, and monitoring
8. Secure coding checklist for developers
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1. What is SQL Injection?

SQL Injection is a class of security vulnerability where untrusted input affects the structure or execution of SQL queries. When input is improperly handled, an attacker could cause the database to run unintended commands or reveal data.

The core problem: Untrusted data mixed with query structure.

---

2. Why it matters (risks & impact)

SQL injection can lead to serious consequences:

- 🔓 **Data leakage**: Personal or sensitive data exposure
- 🗑️ **Data modification or deletion**: Unauthorized changes to critical information
- 🚪 **Authentication bypass**: Circumventing login mechanisms
- 💻 **Complete system compromise**: If database credentials or OS-level commands are exposed
- ⚖️ **Legal, financial, and reputational consequences**: Regulatory fines and brand damage

---

3. High-level types of SQLi (conceptual)

Understanding these categories helps defenders know what to look for in logs and responses:

In-band SQLi

The same channel is used to both inject and get results (e.g., error messages).

Blind SQLi

Attacker learns true/false or timing information when responses don't directly reveal data.

Out-of-band SQLi

Uses alternate channels to retrieve results (rare and environment-dependent).

---

4. How SQLi happens — a conceptual example

Imagine an application that builds a query by concatenating user input directly into SQL:

```
-- ⚠ Conceptual (dangerous) pattern:
query = "SELECT * FROM users WHERE username = '" + userInput +
"'";
```

If `userInput` contains special characters or SQL fragments, it can change the intended query structure.

Key idea: The vulnerability isn't the database — it's how the application constructs the query using untrusted input.

---

5. Vulnerable vs. secure code (safe examples)

❌ Vulnerable pattern (conceptual demonstration)

```
// Do NOT use: concatenating raw user input into SQL
username = getRequestParam("username")
sql = "SELECT * FROM users WHERE username = '" + username +
"'"
db.execute(sql)
```

This shows the pattern that causes risk, but does not provide exploit strings.

✅ Secure pattern — Parameterized queries / prepared statements

Use parameters so input never changes query structure:

```
// ✅ Preferred: parameterized query
username = getRequestParam("username")
sql = "SELECT * FROM users WHERE username = ?"
db.execute(sql, [username])   // driver sends query & data
separately
```

✅ Additional secure practice — ORM / query builders

Using a well-maintained ORM or query builder (correctly) typically avoids manual string building, but still requires care (e.g., avoid raw SQL APIs unless necessary).

Examples:

```
# Python with SQLAlchemy ORM
user = session.query(User).filter(User.username ==
username).first()
```

```
// Node.js with Sequelize ORM
const user = await User.findOne({ where: { username: username
} });
```

```
// Java with JPA/Hibernate
User user = entityManager.createQuery(
    "SELECT u FROM User u WHERE u.username = :username",
User.class)
    .setParameter("username", username)
    .getSingleResult();
```

6. Top prevention techniques (practical, prioritized)

✅ Primary Defenses

1. **Use parameterized queries / prepared statements everywhere**
    - Most effective single defense
    - Separates code from data completely
2. **Use stored procedures with parameters**
    - Only if they use parameters and don't build SQL strings unsafely
    - Centralized business logic
3. **Use least privilege database accounts**
    - App account should only have needed permissions
    - Separate accounts for different application functions

✅ Secondary Defenses

4. **Input validation and normalization**
    - Enforce allowed formats (e.g., email regex, numeric ranges)
    - Use as safety net, not primary defense
5. **Proper escaping**

- Only when parameterization is not possible
- Must use the DB driver's proper escaping functions

6. **Use allowlist (whitelisting) for critical parameters**
   - Especially for sort column names, table names
   - Validate against known good values

Supporting Defenses

7. **Use ORMs/query builders carefully**
   - Avoid raw query concatenation
   - Understand ORM's SQL generation
8. **Centralize DB access logic**
   - Reduce code duplication and mistakes
   - Easier to audit and maintain
9. **Web Application Firewalls (WAFs)**
   - Additional layer, not replacement for secure code
   - Can detect and block common patterns
10. **Keep everything updated**
    - Patch drivers, ORMs, and frameworks regularly
    - Monitor security advisories

---

7. Detection, logging, and monitoring

Logging Strategy

- Log failed queries and unusual DB errors
- ⚠ Be careful not to log sensitive data
- Monitor query patterns and latency spikes
- Look for unusual application errors or stack traces

Monitoring Indicators

- Anomalies that can indicate probing attempts
- High volume of failed parameterized queries
- Unexpected error types or frequencies
- Unusual database access patterns

Testing Tools

- **SAST (Static Application Security Testing)**: During development
- **DAST (Dynamic Application Security Testing)**: In testing environments
- **Database activity monitoring**: Production environments

8. Secure coding checklist (practical, copyable)

✅ Development Checklist

- ☐ All DB queries use parameterized queries or prepared statements
- ☐ No raw concatenation of user input into SQL strings
- ☐ Inputs validated with allowlists for format and length
- ☐ App DB user has minimal privileges (no DROP/ALTER unless required)
- ☐ Error messages returned to users are generic
- ☐ Detailed errors logged only server-side
- ☐ Sensitive data in logs is redacted

✅ Deployment Checklist

- ☐ Regular dependency and runtime patching is scheduled
- ☐ Security tests run in CI (SAST/DAST) before production deploy
- ☐ Production WAF or equivalent protections in place
- ☐ Incident response plan includes data breach steps

---

9. Responsible testing and legal/ethical notes

⚖ Legal Considerations

⚠ **Warning**: Never test for SQLi on systems you don't own or have explicit permission to test. Unauthorized security testing is illegal in many jurisdictions.

□ Safe Testing Practices

- □ Use staging/test environments that mirror production
- □ Follow responsible disclosure process
- □ Coordinate with application owners
- □ Use automated tools for defensive remediation only

□ Recommended Testing Environments

- Local VMs designed for learning
- Intentionally vulnerable applications you own
- Dedicated security testing labs
- Authorized penetration testing environments

---

10. Learning resources & next steps

Essential Reading

- **OWASP Top Ten** — Read the Injection category and remediation guidance
- **OWASP SQL Injection Cheat Sheet** — Defensive best practices
- **Secure coding guides** for your language/framework

Platform-Specific Resources

Python

- Python DB-API 2.0 - Parameterized queries
- SQLAlchemy Security

Java

- JDBC PreparedStatement
- Hibernate Security Guide

Node.js

- node-postgres Parameterized Queries
- Sequelize Security

PHP

- PDO Prepared Statements
- mysqli Prepared Statements

🏃‍♂️ Next Steps

1. **Learn to configure secure DB drivers** and connection libraries
2. **Practice secure coding** in controlled lab environments
3. **Set up automated security testing** in your CI/CD pipeline
4. **Implement monitoring and alerting** for your applications
5. **Create incident response procedures** for security events

---

Quick Takeaway

Don't concatenate. Parameterize. That's the shortest path to preventing SQL injection.

Combine parameterization with:

- Least privilege
- Input allowlists
- Proper logging
- Regular testing

...to build a secure application.

Additional Resources

- **OWASP**: https://owasp.org
- **SANS Secure Coding**: https://www.sans.org/white-papers/
- **NIST Cybersecurity Framework**: https://www.nist.gov/cyberframework

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Last updated: October 2025
This guide focuses on defensive practices and responsible security education.